

Academic Curriculum Vitae

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Chair of AI Systems Engineering, University of Bamberg
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September 17, 2026

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1 Contact Information

University of Bamberg

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2 Online Profiles

Personal homepage: <http://christoph-benzmueller.de/>
Bamberg AISE group: <https://www.uni-bamberg.de/aise/>
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Mathematics Genealogy Project: <https://www.mathgenealogy.org/id.php?id=173483>
Google Scholar: <https://scholar.google.de/citations?user=zD0vtfwAAAAJ&hl=de>
ResearchGate: <https://www.researchgate.net/profile/Christoph-Benzmueller>
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3 Curriculum Vitae

3.1 Education

02/2012	Freie Universität Berlin, <i>venia legendi</i> (Mathematics).
01/2001–07/2007	Saarland University, postdoctoral studies; habilitation / <i>venia legendi</i> (Computer Science).
04/1995–07/1999	Saarland University, doctoral studies; <i>Dr.-Ing.</i> (AI & Computer Science).
10/1991–03/1995	Saarland University, diploma studies (<i>Diplom</i> , Computer Science & Economics).
10/1989–10/1991	Saarland University, preliminary diploma (<i>Vordiplom</i> , Computer Science & Economics).
08/1985–06/1988	Auguste-Viktoria-Gymnasium Trier, 13th-grade completion (<i>Allgemeine Hochschulreife</i>).
08/1979–07/1985	Gymnasium Saarburg, 10th-grade completion (<i>Sekundarabschluss I</i>).
08/1975–07/1979	Grundschule St. Marien, Saarburg, 4th-grade completion.

3.2 Diploma Thesis, Doctoral Thesis, Habilitation Script

Diploma Thesis (1994) – *Eine Fallstudie zur Spezifikation von Systemanforderungen in der Spezifikationssprache OBSCURE*, 194 pages, Saarland University. <http://christoph-benzmueller.de/papers/T1.pdf>

Advisors: Jacques Loeckx (Saarland University) and Klaus Gersonde (Fraunhofer-Institut Biomedizinische Technik, IBMT, St. Ingbert).

Synopsis: Executable formal specification of a knowledge-based information, visualization and maintenance system in bioinformatics; the diploma project was conducted at the Fraunhofer-Institut für Biomedizinische Technik in St. Ingbert.

Doctoral Thesis (1999) – *Equality and Extensionality in Automated Higher-Order Theorem Proving*, 178 pages, Saarland University. <http://christoph-benzmueller.de/papers/T2.pdf>

Advisors: Jörg Siekmann (Saarland University), Michael Kohlhase (Saarland University, now FAU Erlangen-Nürnberg), Frank Pfenning (Carnegie Mellon University, USA); *mentor:* Peter Andrews (Carnegie Mellon University, USA).

Synopsis: Study of equality and extensionality in higher-order logic with novel contributions to the proof theory, model theory and automation of classical higher-order logic. The thesis provided the theoretical foundation of the LEO theorem provers and initiated the development of an international infrastructure for higher-order automated theorem proving; I have, following Peter Andrews and his group at CMU, significantly contributed to the pioneering of this area; see (35, 28) for a discussion of the historical developments.

Habilitation (2006) – *Cumulative Habilitation Script (on Integrated Mathematics Assistance Systems and the Automation of Higher-Order Logic)*, 357 pages, Saarland University. <http://christoph-benzmueller.de/papers/T3.pdf>

Synopsis: Argument for the development of integrated assistance systems for formal proof developments in mathematics; own contributions with a focus on the integration of AI technologies (planning, agent-based paradigms, natural-language tutoring, learning and abstraction techniques, mathematical ontologies, knowledge graphs). The foreword of the habilitation script, published as a separate journal article *Towards Computer Aided Mathematics* (63), sketches some main challenges of the area.

3.3 Positions, Affiliations and Work Contracts

since 02/2022	Full Professor (Chair), AI Systems Engineering, University of Bamberg.
since 08/2021	Adjunct (<i>apl.</i>) Professor, Dep. of Mathematics and Computer Science, Freie Universität Berlin.
since 10/2020	Consultant and scientific advisor (freelancer), Berlin / Bamberg.
10/2023–03/2024	Visiting Scholar / Guest Professor, BITS Pilani Dubai, UAE.
04/2021–04/2022	Consultant and scientific advisor, IDNI AG, Liechtenstein.
10/2020–04/2021	Consultant and scientific advisor, Latentine GmbH, Berlin.
04/2020–09/2020	First <i>Una Europa</i> Chair, Dep. of Mathematics and Computer Science, Freie Universität Berlin.
09/2018–12/2018	Consultant and scientific advisor, BILLON Sp. z o.o., Warsaw, Poland.
07/2018–03/2020	Guest Professor, Dep. of Mathematics and Computer Science, Freie Universität Berlin.
03/2017–06/2019	Visiting Scholar, AI and Computer Science, University of Luxembourg.
09/2015–08/2016	Visiting Scholar, CSLI, Philosophy, Stanford University, USA.
02/2012–01/2017	DFG Heisenberg Fellow, Dep. of Mathematics and Computer Science, Freie Universität Berlin.
since 02/2012	Privatdozent, <i>venia legendi</i> in Mathematics, Freie Universität Berlin.
05/2011–11/2011	DFG return scholarship, Dep. of Mathematics and Computer Science, Freie Universität Berlin.
12/2008–12/2009	Full Professor, Formal Methods and AI, International University in Germany, Bruchsal.
11/2008–04/2011	Senior Researcher, Semantic Technologies, Articulate Software, Angwin, CA, USA.
since 07/2007	Privatdozent, <i>venia legendi</i> in Computer Science, Saarland University.
10/2006–10/2007	Senior Research Fellow, The Computer Laboratory, University of Cambridge, UK.
01/2007–09/2007	Research Assistant, DFKI GmbH, Saarbrücken.

03/2004–10/2008 University Lecturer, Computer Science, Saarland University; civil servant (C2-grade, granted until 2012).

01/2001–02/2004 University Lecturer, Computer Science, Saarland University; civil servant (C1-grade).

01/2000–12/2000 Postdoctoral Research Fellow, Computer Science, Universities of Birmingham & Edinburgh, UK.

09/1999–12/2000 Postdoctoral Research Fellow, Graduate College Cognitive Science, Saarland University.

10/1996–04/1997 Short-term Scholar, Department of Mathematics, Carnegie Mellon University, Pittsburgh, USA.

04/1995–08/1999 Research Assistant (BAT IIa), Computer Science, Saarland University.

08/1993–07/1994 External diploma project, Fraunhofer-Institut Biomedizinische Technik, St. Ingbert.

01/1997–06/1997 Research Assistant, DFKI GmbH, Saarbrücken.

10/1992–01/1995 Research Assistant (*wiss. Hilfskraft*), DFKI GmbH, Saarbrücken.

10/1991–04/1994 Research Assistant (*wiss. Hilfskraft*), Saarland University & industry, BMFT project KORSO.

3.4 Career Breaks

2009–2010 12 months parental leave.

2011–2012 3 months parental leave + 2 months care leave.

3.5 Awards and Honors

Best Paper Awards: [DEON](#) conference 2018 (John-Jules Meyer Best Paper Award); [CLAR](#) conference 2023; [ISDIA](#) conference 2025.

Director (elected), [Centre for Innovative Applications of Computer Science \(ZIAI\)](#), University of Bamberg, since 2023.

[High-end Foreign Expert](#) for Machine Ethics in cross-cultural context, Zhejiang University, China, since 2023.

Cover story on my research in computational metaphysics in the science magazine [Science & Vie](#), 2020.

National Contact Point, Confederation of Laboratories for AI Research in Europe ([CAIRNE / CLAIRE](#)), since 2019.

World Champion in Higher-Order Automated Theorem Proving (LTB division), CASC 2019, with the LEO-III prover.

DFG mentor for the 30th European Union Contest for Young Scientists (EUCYS) in Dublin, 2018.

Volkswagen Stiftung *Experiment!* programme, project *CRAP: Consistent Rational Argumentation in Politics*, selected for funding (17 of 594 submissions; $\leq 3\%$ acceptance rate), 2017.

FU Berlin Central Teaching Award for the lecture course *Computational Metaphysics*, 2015.

Invited Visiting Scholar (10 months), Stanford University, 2015/2016.

Nominated for the *Dan David Prize* in the category *Future: Artificial Intelligence – Digital Minds*, Israel, 2014.

Shortlisted for the *Amalia Preis für Neues Denken*, Weimar, Germany, 2013.

Heisenberg Fellowship of the Deutsche Forschungsgemeinschaft (DFG), 2012–2017.

Faculty member of the [Berlin Mathematical School](#), since 2012.

Venia legendi (Lehrbefugnis) in Mathematics since 2012, FU Berlin.

World Champion in Higher-Order Automated Theorem Proving, CASC 2010, with the LEO-II prover.

Venia legendi (Lehrbefugnis) in Computer Science since 2007, Saarland University.

Invited Visiting Scholar (12 months), University of Cambridge, UK, 2006/2007.

Member of the DFG Collaborative Research Center SFB 378, 2001–2008.

Postdoctoral Research Fellow of the Graduate College Cognitive Science, Saarland University, 1999–2000.
Invited Scholar at the Department of Mathematics, Carnegie Mellon University (invitation by P. Andrews), 1996/1997.

PhD scholarship of the [Studienstiftung des Deutschen Volkes](#), 1996–1998.

Awards for students supervised by me (selection)

2023 Daniel Kirchner – shortlisted as *KI-Newcomer* in an initiative by GI and BMBF.
2020 Veronika Solopova – Elsa-Neumann Scholarship for PhD students, State of Berlin.
2019 Alexander Steen – nominated for the *Ernst-Reuter-Preis*, FU Berlin.
2018 Alexander Steen – GI Junior Fellowship.
2018 David Fuenmayor – FUBright mobility grant, FU Berlin.
2018 Hanna Lachnitt – Helmut-Veith Award, TU Wien, for a BSc thesis under my supervision at FU Berlin.

Various travel awards and stipends for further students.

4 Service to the Academic Community

4.1 Commissions of Trust

since 2026 Jury member, Herbrand Award.
since 2022 Jury member, Dov Gabbay Prize for Logic and Foundations.
since 2019 Jury member, Kurt Gödel Essay Award (Kurt-Gödel-Freundeskreis Berlin).

Reviewer for funding agencies and universities

- Alexander von Humboldt Stiftung, DE
- Austrian Science Fund (FWF), AT
- Cambridge University, GB
- Czech Science Foundation, CZ
- Deutsche Forschungsgemeinschaft (DFG), DE
- Fulbright Germany, DE
- Johannes Kepler Universität, Linz, AT
- King's College, London, GB
- Natural Sciences and Engineering Research Council of Canada (NSERC), CA
- Recherche en sciences & technologies de l'information (DIGITEO), FR
- Swiss National Science Foundation (SNSF), CH
- University of Bonn, DE

4.2 Editorial Board Memberships

2025–ongoing Associate Editor, *Annals of Mathematics and Artificial Intelligence*.
2021–ongoing Editorial Board, *Mathematics in Computer Science*, Springer.
2020–ongoing Editorial Board, *Historia Logicae*, College Publications.
2018–ongoing Editorial Board, *Journal of Applied Logics – IfCoLoG Journal of Logics and their Applications*.
2008–ongoing Editorial Board, *Logic Journal of the IGPL*, Oxford University Press.
2008–2018 Editorial Board, *Journal of Applied Logic*, Elsevier.

4.3 Scientific Society Memberships (past and current)

<i>since 2024</i>	Sigma Xi, USA (elected associate member).
<i>2023–2024</i>	Conference on Automated Deduction (CADE), as IJCAR co-chair.
<i>since 2022</i>	Bamberg Centre for Artificial Intelligence (BaCAI), U Bamberg, core team.
<i>since 2022</i>	Centre for Innovative Applications of Computer Science (ZIAI), U Bamberg; director since 2023.
<i>since 2022</i>	Federation of German Scientists (Vereinigung Deutscher Wissenschaftler, VDW).
<i>since 2020</i>	International AIQT Foundation (AI and Quantum Technology), Switzerland; charter member.
<i>2020–2021</i>	World Congress on Logic and Religion, scientific committee.
<i>2020–2021</i>	Conference on Intelligent Computer Mathematics (CICM), steering committee.
<i>since 2019</i>	Fachbereich Künstliche Intelligenz, Gesellschaft für Informatik (GI), steering committee.
<i>since 2019</i>	Confederation of Laboratories for AI Research in Europe (CLAIRE), key supporter.
<i>since 2018</i>	Society of Deontic Logic and Normative Systems (DEON), elected steering committee member.
<i>since 2018</i>	Kurt-Gödel-Freundeskreis Berlin.
<i>since 2018</i>	Berlin Mathematical School (BMS+) / Berlin Mathematics Research Center MATH+.
<i>2015–2023</i>	Association for Automated Reasoning (AAR), board member.
<i>2015–2021</i>	Conference on Automated Deduction (CADE), elected member; vice-president.
<i>2014–2018</i>	Spokesman of the Section <i>Deduction Systems</i> of the Gesellschaft für Informatik (GI).
<i>2014–2018</i>	Fachbereich Künstliche Intelligenz, GI, steering committee.
<i>2014–2016</i>	Berlin Mathematical School (BMS), mentoring, gender & diversity committee.
<i>2010–2018</i>	User Interfaces for Theorem Provers, steering committee.
<i>since 2010</i>	International Federation for Computational Logic (IfCoLoG), executive board.
<i>2008–2011</i>	Conference on Automated Deduction (CADE), elected trustee.

General society memberships: Gesellschaft für Informatik; Deutscher Hochschulverband; Association for Automated Reasoning; American Association for Artificial Intelligence (AAAI); Deutsche Vereinigung für Mathematische Logik und für Grundlagenforschung der Exakten Wissenschaften; Kurt Gödel Society, Vienna; Scandinavian Logic Society.

4.4 Advisory Board Memberships

<i>since 2025</i>	Bamberg Mathematical Consciousness Science Initiative (BAM Ξ).
<i>since 2020</i>	International AIQT Foundation (AI and Quantum Technology), Switzerland.
<i>2021–2022</i>	IDNI AG, Liechtenstein.
<i>2016–2018</i>	ARD: Accessible Reasoning with Diagrams, Leverhulme Trust, Cambridge U and U Brighton, GB.

4.5 Programme Committees and Reviewing for Journals

4.5.1 Reviewing for journals

- ACM Transactions on Computational Logic (ACM TOCL)
- AI & Society
- AI Communications (AICom)
- Analysis
- Annals of Mathematics and Artificial Intelligence (AMAI)

- Australasian Journal of Logic
- Digital Society
- Erkenntnis (ERKE)
- Fundamenta Informaticae
- Historia Logicae
- IEEE Transactions on Artificial Intelligence
- International Journal of Semantic Web and Information Systems (IJSWIS)
- International Journal of Software and Informatics (IJSI)
- Journal of Applied Logics – IfCoLoG Journal of Logics and their Applications
- Journal of Applied Non-Classical Logics (JANCL)
- Journal of Automated Reasoning (JAR)
- Journal of Computation and Mathematics (LMS)
- Journal of Logic and Computation (JLC)
- Journal of Philosophical Logic (JPL)
- Journal of Symbolic Computation (JSC)
- Journal of Web Semantics (JWS)
- Künstliche Intelligenz (KI)
- Large-Scale Data- and Knowledge-Centered Systems
- Logica Universalis
- Logical Methods in Computer Science (LMCS)
- Mathematics in Computer Science (MCS)
- Review of Symbolic Logic (RSL)
- S+F Sicherheit und Frieden
- Studia Logica
- Synthese
- TheoLogica: An International Journal for Philosophy of Religion and Philosophical Theology
- Theology and Science
- Zentralblatt MATH

4.5.2 Programme committee memberships – conferences (past and current)

- Conference on Agents and Artificial Intelligence (ICAART)
- Conference on Artificial Intelligence and Applications (AIA)
- Conference on Artificial Intelligence and Symbolic Computation (AISC)
- Conference on Automated Deduction (CADE)
- Conference on Automated Reasoning (IJCAR)
- Conference on Automated Reasoning with Analytic Tableaux and Related Methods (TABLEAUX)
- Conference on Formal Ontology in Information Systems (FOIS)
- Conference on Frontiers of Combining Systems (FroCoS)
- Conference on Information Security (IFIP SEC)
- Conference on Intelligent Computer Mathematics (CICM)
- Conference on Principles of Knowledge Representation and Reasoning (KR)
- Conference on Logic for Programming, AI and Reasoning (LPAR)
- Conference on Mathematical Knowledge Management (MKM)
- Conference on Non-Classical Logics: Theory and Applications (NCL)
- Conference on Theorem Proving in Higher Order Logics (TPHOLs)
- Digital Humanism – Interdisciplinary Science and Research Conference (DIGHUM)
- European Summer School in Logic, Language and Information (ESSLLI), area co-chair logic & compu-

tation

- European Summer School on Artificial Intelligence (ESSAI)
- German Conference on Artificial Intelligence (KI)
- Global Conference on Artificial Intelligence (GCAI)
- International Conference on Artificial Intelligence and Law (ICAIL)
- International Conference on Artificial Intelligence Logic and Applications (AILA)
- International Conference on Computer Science, Applied Mathematics and Applications (ICCSAMA)
- International Conference on Deontic Logic and Normative Systems (DEON)
- International Conference on Interactive Theorem Proving (ITP)
- International Conference on Knowledge Engineering and Ontology Development (KEOD)
- International Conference on Logic and Argumentation (CLAR)
- International Conference on Logic Programming (ICLP)
- International Joint Conference on Artificial Intelligence (IJCAI)
- Intl. Conf. on Ontologies, Databases, and Applications of Semantics for Large Scale Information Systems (ODBASE)
- Symposium on Moral and Legal AI Alignment
- Symposium on Symbolic Computation and Mechanized Reasoning (CALCULEMUS)
- Symposium on Values & Preferences in Reasoning (VPR)

4.5.3 Programme committee memberships – workshops and others

- International ARCADE Workshop on Automated Reasoning
- International Workshop on Artificial Intelligence and Ethics
- International Workshop on AI Compliance Mechanism (WAICOM)
- International Workshop on Automated Reasoning about Context and Ontology Evolution (ARCOE)
- International Workshop on Automated Reasoning with Connection Calculi (AReCCa)
- International Workshop on Logics for New-Generation Artificial Intelligence (LNGAI)
- Joint Automated Reasoning Workshop and Deduktionstreffen (ARWDT)
- Meeting of the German Interest Group on Deduction Systems (Deduktionstreffen)
- The ATP System Competition (CASC)
- TPTP World Workshop (TPTPWöWo)
- Workshop on Automated Theorem Proving meets Interactive Theorem Proving (AMI)
- Workshop on Enabling Domain Experts to use Formalised Reasoning (Do-Form)
- Workshop on Acquisition, Representation and Reasoning with Contextualized Knowledge (ARCOE)
- Workshop on Agents and Automated Reasoning at IJCAI
- Workshop on Comparative Empirical Evaluation of Reasoning Systems (COMPARE)
- Workshop on Evaluation Methods for Solvers and Quality Metrics for Solutions (EMS+QMS)
- Workshop on Formal and Cognitive Reasoning (FCR)
- Workshop on Future Directions in Automated Reasoning at IJCAR
- Workshop on Inference in Computational Semantics (ICoS)
- Workshop on Knowledge Exchange: Automated Provers and Proof Assistants (KEAPPA)
- Workshop on Logic for Automated Reasoning and Automated Reasoning for Logic (LARARL)
- Workshop on Mathematical Intelligent Proof Search (MIPS)
- Workshop on Models of Legal Reasoning
- Workshop on Moral and Legal AI Alignment
- Workshop on Practical Aspects of Automated Reasoning (PAAR)
- Workshop on Proof Exchange for Theorem Proving (PxTP)
- Workshop on the Implementation of Logics (IWIL)

- Workshop on Unification (UNIF)
- Workshop on User Interfaces for Theorem Provers (UITP)

4.5.4 Additional reviewing for conferences

- Conference on Automated Deduction (CADE)
- Conference on Automated Reasoning (IJCAR)
- Conference on Computational Logic (CL)
- Conference on Foundations of Software Science and Computation Structures (FoSSaCS)
- Conference on Logic for Programming, Artificial Intelligence and Reasoning (LPAR)
- Conference on Logics in Artificial Intelligence (JELIA)
- Conference on Types for Proofs and Programs (TYPES)
- European Conference on Artificial Intelligence (ECAI)
- German Conference on Artificial Intelligence (KI)
- IEEE International Symposium on Multiple-Valued Logic (ISMVL)
- Symposium on Logic in Computer Science (LICS)
- Symposium on Symbolic and Algebraic Computation (ISSAC)
- Florida Artificial Intelligence Research Society Conference (FLAIRS)
- Various other conferences and workshops not listed here.

4.6 Invitations to Dagstuhl Seminars and Other Invitational Meetings

Schloss Dagstuhl is one of the world's premier meeting centres for research in computer science and artificial intelligence; participation in a Dagstuhl Seminar is by personal invitation only, based on scientific reputation.

2026	Dagstuhl Seminar 26091, <i>Revisiting the Foundations of Deduction in a New World.</i>
2023	Dagstuhl Seminar 23401, <i>Automated Mathematics: Integrating Proofs, Algorithms and Data.</i>
2023	Dagstuhl Seminar 23371, <i>Roadmap for Responsible Robotics.</i>
2023	Dagstuhl Seminar 23151, <i>Normative Reasoning for AI.</i>
2021	Dagstuhl Seminar 21371, <i>Integrated Deduction</i> (could not attend).
2019	Dagstuhl Seminar 19171, <i>Ethics and Trust: Principles, Verification and Validation</i> (could not attend).
2018	Dagstuhl Seminar 18171, <i>Normative Multi-Agent Systems.</i>
2017	Dagstuhl Seminar 17371, <i>Deduction Beyond First-Order Logic.</i>
2016	Dagstuhl Seminar 16421, <i>Universality of Proofs.</i>
2015	Dagstuhl Seminar 15381, <i>Information from Deduction: Models and Proofs.</i>
2010	Dagstuhl Seminar 10412, <i>QSTRLib: A Benchmark Problem Repository for Qualitative Spatial and Temporal Reasoning.</i>
2005	Dagstuhl Seminar 05431, <i>Deduction and Applications.</i>
2021	DFG–NSF Research Workshop on <i>Cybersecurity and Machine Learning.</i>
2019	DFG Tagung <i>Traceability and Securing of Results.</i>
2018	DFG Diskussionstreffen <i>Bedeutung und Struktur der Forschung an digitalen Methoden und Verfahren?</i>
2017	DFG Heisenberg-Vernetzungstreffen.
2015	DFG Rundengespräch zum Schwerpunktprogramm <i>Argument Machines.</i>

4.7 Organisation of Scientific Events

2025	Winners' Symposium, 2025 Kurt Gödel Essay Prize (Kurt-Gödel-Freundeskreis Berlin), Bamberg – co-organizer (https://tinyurl.com/54w5hvsj).
2025	International Workshop on Computational Ontology & Metaphysics, Bamberg – programme chair and co-organizer (https://tinyurl.com/54w5hvsj).
2024	International Joint Conference on Automated Reasoning (IJCAR) – programme co-chair.
since 2014	Automated Reasoning in Quantified Non-Classical Logics (ARQNL) – co-organiser / co-chair.
2022	Goethe-Institut Summer School on AI and Ethics – local co-organiser at U Bamberg and programme advisor.
2022	European Summer School in Logic, Language and Information (ESSLLI) – area co-chair.
2021	International Conference on Logic and Argumentation (CLAR) – programme co-chair.
2020	Conference on Intelligent Computer Mathematics (CICM) – programme co-chair.
2019	German Conference on Artificial Intelligence (KI) – programme co-chair.
2019	International Workshop <i>Kurt Gödel: Philosophical Views</i> , Berlin – organiser.
2018	AI: Truth or Dare, round-table discussion with KPMG Luxembourg – co-organiser.
2018	Luxembourg Logic for AI Summit (LuxLogAI) – initiation, conception and co-organisation.
2018	International Joint Conference on Rules and Reasoning (RuleML+RR) – programme co-chair.
2017	Global Conference on Artificial Intelligence (GCAI) – programme co-chair.
2016	Global Conference on Artificial Intelligence (GCAI) – programme co-chair and conference chair.
2015	Conference on Automated Deduction (CADE-25) – conference chair of the 25th jubilee edition.
2015–2017	Meeting of the German Interest Group on Deduction Systems – co-organiser (2015, 2016, 2017).
2006–2014	Workshop on User Interfaces for Theorem Provers (UITP) – co-organiser / co-chair (2006, 2008, 2014).
2013	Conference on Automated Deduction (CADE) – workshop and competition chair.
2008	Conference on Intelligent Computer Mathematics (CICM) – doctoral program.
2008	Workshop on Evaluation of Systems for Higher-Order Logic (ESHOL) – co-organiser / co-chair.
2007	Conference on Automated Deduction (CADE) – workshop and tutorial chair.
2006	Workshop on the Implementation of Logics (IWIL) – co-organiser / co-chair.
2006	Workshop on Tutorial NL Dialog in Mathematics (MathDialog) – co-organiser / co-chair.
2005	Workshop on Empirically Successful Automated Higher-Order Reasoning (ESHOL) – co-organiser / co-chair.
2005	Workshop on Mathematical Proof Assistants (Theorema-Ultra-Omega) – co-organiser / co-chair.
2005	Workshop on Higher-Order Logics and Hybrid Logics, Saarbrücken-Nancy – co-organiser / co-chair.
2004	Workshop on Computer-Supported Mathematical Theory Development – co-organiser / co-chair.
2004	Evaluation of the Collaborative Research Center SFB 378 – co-organiser.
2004	Meeting of the German Interest Groups on Deduction Systems & Logics in CS – co-organiser.

2003	Workshop CALCULEMUS at the Symposium on Mathematics on the Semantic Web – organiser.
2003	Midterm Review of the EU CALCULEMUS Research Training Network – organiser / chair.
2002	Autumn School of the EU CALCULEMUS Research Training Network in Pisa – co-organiser.
2000	Meeting of the German Interest Group on Deduction Systems.

4.8 Administrative Academic Services

since 2023	Quality-circle member of various BSc study programmes at U Bamberg.
since 2023	Head of the BSc programme in AI and Data Science at U Bamberg.
since 2023	Study Programme Advisor of the BSc programme in AI and Data Science at U Bamberg.
since 2023	Member of the Examination Board for Computer Science and Applied Computer Science at U Bamberg.
2014–2015	Chair of appointment committee for the W1 professorship in Mobile Robotics, FU Berlin.
2015	Member of appointment committee for the W3 professorship in Bioinformatics and Informatics, FU Berlin.
2014–2015	Member of appointment committee for the W1 professorship in AI, FU Berlin.

4.9 Social Commitment and Community Involvement

since 2022	Active member of the Federation of German Scientists (VDW).
since 2023	Supporter of the initiative <i>Quo Vadis KI?</i> (https://ki-folgen.de/).
since 2021	Supporter of the initiative against <i>Atomkrieg aus Versehen</i> (https://atomkrieg-aus-versehen.de/).
2019	Organisation of <i>Autonomous Weapon Systems – Dangers and Need for a Ban</i> at the KI 2019 conference.
2019	Organisation of an AI & Robotics presentation day for Kronach-Grundschule (elementary school) at FU Berlin.
2019	Panelist on “ <i>Laws in LAWS</i> ”, Future of Life Institute (FLI) , Berlin.
2018	DFG discussion meeting <i>Bedeutung und Struktur der Forschung an digitalen Methoden und Verfahren?</i>
2016	Invited panel <i>Erwachen der Roboter</i> , Bundeszentrale für politische Bildung, Berlin.
2015	Invitation to the parliamentary breakfast (German Bundestag) on <i>Lethal Autonomous Weapon Systems</i> , Berlin.
2015	Invited participant, <i>Risks of Autonomous Weapon Systems</i> , Friedrich-Ebert-Stiftung, Berlin.
2013–2015	Parent representative, Kindergarten Roonstr., Berlin.
since 1999	Alumnus of the Studienstiftung des Deutschen Volkes.

4.10 Other Activities

since 2024	Top-10 positions in my age group at the German championships in middle-distance running.
1989–1992	Member / athlete of the German Olympic Centre in Saarbrücken, Germany.
1990	German national champion, long-distance running, cross-country team.
1989	Best German participant in the Military Cross-Country World Championships, Tunis, Tunisia.
10/1988–12/1989	Basic military service in sports supporting group (<i>Sportsoldat</i>), Warendorf, Germany.
1988/1989	Elected ombudsman of company, basic military service.

5 System Development (small selection)

2024–ongoing	HOL Tableaux – framework for tableaux-based higher-order theorem proving and unification <i>Languages:</i> Elixir <i>Role:</i> supervisor.
2024–ongoing	Distributed Hybrid AI – distributed, agent-based framework for ATP and ML <i>Languages:</i> Elixir <i>Role:</i> supervisor.
2018–ongoing	LogiKey – framework, methodology and tool support for ethical and legal reasoning <i>Languages:</i> Isabelle/HOL, TPTP THF, Scala <i>Role:</i> project inventor, supervisor, project partner. <i>Comments:</i> Collaboration partners include U Luxembourg, Zhejiang U, TU Wien, U Greifswald.
2018–2024	NLP-Empowered Propaganda Detection Tool <i>Languages:</i> various ML tools, Python <i>Role:</i> PhD supervisor of Veronika Solopova (implementer).
2014–ongoing	Leo-III – agent-based ATP for classical higher-order logic (and various non-classical logics) <i>Languages:</i> Scala <i>Role:</i> project supervisor. <i>Comments:</i> World Champion (CASC LTB) 2019; integrated with Isabelle/HOL (release 2018).
2014–2019	LeoPARD – agent-based platform for the development of knowledge-based reasoning systems <i>Languages:</i> Scala <i>Role:</i> project leader, supervisor.
2007–2014	LEO-II – automated theorem prover for classical higher-order logic (and various non-classical logics) <i>Languages:</i> OCaml <i>Role:</i> project leader, supervisor, main developer. <i>Comments:</i> World Champion (CASC THF) 2010.
2000–2008	DIALOG – demonstrator system on tutorial NL dialog on proofs, integrated with OMEGA <i>Languages:</i> various <i>Role:</i> project leader, supervisor.
1998–2006	OANTS – agent-oriented reasoning system, integration platform for OMEGA <i>Languages:</i> Lisp <i>Role:</i> project leader, co-supervisor and co-developer.
1995–2005	LEO – automated theorem prover for classical higher-order logic <i>Languages:</i> Lisp <i>Role:</i> supervisor and main developer.
1992–2006	OMEGA – large AI-based interactive proof assistant & automated proof planner, NL techniques <i>Languages:</i> Lisp, Oz <i>Role:</i> developer, later project co-leader and co-supervisor of a team of approx. 8–10 developers.

I also initiated and supervised the development of a Conference Management Tool for CADE-25 at FU Berlin by Benjamin Vetter. The development was further supported by Alexander Steen, Max Wisniewski and Julian Röder. The tool was very successful and was subsequently applied to various other conferences at FU Berlin.

6 Media Reports (small selection)

6.1 YouTube and other movie channels

2024	Comments on the formalisation and automation of foundational theories. [video]
2024	AI Consciousness – Notes from a Logician’s Perspective. [video]
2023	Ethik & Künstliche Intelligenz. [video]
2023	ChatGPT, KI & Gott. [video]
2022	What is AI? [video]
2022	Trusted AI through Ethico-Legal Governors. [video]
2022	Logico-pluralistic exploration of foundational theories with computers. [video]
2022	Humanities and the Rise of AI – The Computer as Experimental Metaphysician? [video]
2022	Ontology Talk: An Interview with Prof. Christoph Benz Müller – Parts 1–3. [part 1] [part 2] [part 3]
2021	Classical Higher-Order Logic (HOL) – Automation and Selected Applications. [video]
2020	Una Europa Talk – <i>Resistance is Futile. You Will be Digitalized!</i> [video]
2019	<i>Der Ontologische Gottesbeweis</i> , 71. Science Slam, SO36, Berlin-Kreuzberg. [video]
2018	A Flexible Reasoning Infrastructure for Normative Reasoning, DEON keynote. [video]
2018	Mathematics and Reality, AISSQ keynote. [video]
2016	OEB 2016 – Plenary Debate. [video]
2015	AISSQ 2015: Experiments in Computational Metaphysics – Gödel’s Proof of God. [video]

6.2 Television (selection)

16.04.2023	ZDF TerraX – <i>Die großen Fragen</i> (approx. min 34:00 in the ZDF mediathek); more than 4 million viewers.
31.01.2020	ZDF <i>heute in Deutschland</i> – Bundestag stimmt über “Killerroboter” ab.
31.01.2020	ZDF <i>Morgenmagazin</i> – Vollautomatisierte “Killerroboter”.
30.01.2020	ZDF <i>heute+</i> – Vollautomatisierte “Killerroboter”.
28.07.2016	3sat <i>scobel</i> – <i>Die Roboter-Rivalen</i> ; expert interview on AI and consciousness.
07.04.2016	National Geographic Television – panel discussion on the occasion of the Germany premiere of <i>Morgan Freeman’s Story of God</i> . [photo 1] [photo 2]
23.03.2015	3sat – <i>D wie Deus ex Machina</i> . [video]

6.3 Interviews in newspapers and magazines (selection)

24.10.2022	Fränkischer Tag – <i>Die neuen Professoren an der Uni Bamberg, Teil 1</i> .
09.12.2021	ZJU Newsroom and various newspapers in China – <i>Zhejiang University and University of Luxembourg inaugurate joint lab to explore interdisciplinary collaboration</i> .
03.07.2021	Tagesspiegel – <i>Vorbild Biologie: Wie funktioniert Künstliche Intelligenz? Wie zuverlässig kann sie sein? Und was ist eigentlich Intelligenz? Ein Gespräch mit Robin Hiesinger und Christoph Benz Müller</i> . [link]
12.12.2019	PUCRS – <i>School of Computer Science’s 40th anniversary celebrations come to a close with lecture by Christoph Benz Müller</i> . [link]
09/2015	Albert, Das Journal der Einstein Stiftung Berlin, No. 1 – <i>G(x) = Gott</i> . [link]
06.02.2014	PC Zoznam, Czech Republic – <i>Interview: Christoph Benz Müller ... the computer then proved: there exists God</i> (English).

28.12.2013	Marabilias, Spain – <i>Demostrado científicamente, Dios existe.</i>
03.11.2013	Tendencias21, Spain – <i>Verificación informática de la existencia de Dios ...y filosofía asistida por ordenadores.</i>
24.09.2013	Motherboard, Germany – <i>Gottesbeweis auf dem MacBook.</i>
26.08.2013	Telepolis, Germany – <i>Computer beweist die Existenz Gottes</i> ; among the ten most-read and most-commented articles on <i>heise.de</i> in 2013. [link]

6.4 Radio interviews and contributions (selection)

25.03.2022	SWR2 – <i>Seele reloaded: Menschliches Bewusstsein zwischen digitalen Clouds und KI.</i> [link]
15.01.2022	Deutschlandfunk – <i>Wenn der Computer entscheidet – Möglichkeiten und Grenzen von KI.</i> [link]
10.05.2017	Deutschlandfunk – <i>Roboterethik: Die Maschinen werden autonom.</i> [link]
2017	Ö1 (ORF) – <i>Radiokolleg: Die Welt begreifen – Warum wir nicht wie Maschinen lernen</i> (Gestaltung: Marianne Unterluggauer); 3 parts, 9–11 Jan 2017. [link]
28.09.2016	Die Evangelische Funkagentur, <i>Hauptsache Mensch – Es gibt einen Gott, sagt der Computer.</i>
10.09.2013	detektor.fm, Germany – <i>Mathematischer Gottesbeweis von Forschern bestätigt.</i>
09.09.2013	ORF.at, Austria – <i>Computer bestätigt Gottesbeweis.</i>

6.5 Magazines, newspaper articles and online media (small selection)

01.03.2024	CHIP – <i>KI im Kampf.</i> [link]
27.04.2024	Vereinigung Deutscher Wissenschaftler, series <i>Jung und Alt bewegt</i> with Prof. Hartmut Graßl – <i>KI Insights: Experte Prof. Dr. Christoph Benz Müller über Ethik, Innovation und die globale Perspektive auf künstliche Intelligenz.</i> [link]
19.02.2023	Der Tagesspiegel – <i>Künstliche Intelligenz: Der plappernde Papagei im Netz.</i> [link]
19.02.2023	FU Berlin – <i>Der Papagei im Netz.</i> [link]
23.03.2022	Universität Bamberg – <i>Künstliche Intelligenz nicht mehr wegzudenken aus unserer Zukunft.</i> [link]
2022	<i>Vertrauenswürdige und Verantwortungsvolle KI für unsere große Stadt</i> ; Science Slam, part of the <i>Wissensstadt Berlin 2021</i> science festival – winner of the contest. [link]
08.10.2020	Die Tagespost – <i>Kann man Gott mathematisch beweisen?</i> [link]
03.08.2020	Science & Vie – <i>Existence de Dieu: les mathématiques ont enfin la réponse.</i> [link]
24.10.2019	Wolfsburger Nachrichten – <i>Wolfsburg: Beim AI-Camp geht es um Künstliche Intelligenz.</i> [link]
25–27.09.2019	Heise.de – four reports from the KI 2019 conference (which I co-chaired). [1] [2] [3] [4]
16.10.2019	Spektrum der Wissenschaft – <i>Ist die Existenz Gottes logisch?</i> [link]
08.09.2019	Spektrum der Wissenschaft – <i>Gibt es eine Logik des Glaubens?</i> [link]
10.09.2019	DWIH São Paulo – <i>Artificial Intelligence Welcomes Humans.</i> [link]
09.09.2019	inar.de – <i>AI Camp: Wie beeinflusst Künstliche Intelligenz unseren Alltag?</i> [link]
04.09.2019	University of Luxembourg – <i>World-championship in Automated Reasoning: computer scientist awarded.</i> [link]
03.07.2019	campus.leben FU Berlin – <i>Von RoboFish und Baybie für die Zukunft lernen.</i> [link]
16.05.2019	Faz.net – <i>Algorithmensicherheit: Von Logik umgebracht oder gerettet.</i> [link]
27.04.2019	Tagesspiegel – <i>Was bedeutet Wissenschaftsfreiheit für Sie?</i> [link]
07.10.2018	orissadiary.com – <i>Odisha Governor inaugurates 11th AISSQ conference on Mathematics and Reality at IIT Bhubaneswar.</i> [link]

05.09.2018 Industry of Things – *KI: Die Dampfmaschine 4.0 fürs 21. Jahrhundert?* [\[link\]](#)

10.05.2017 Deutschlandfunk – *Roboterethik: Die Maschinen werden autonom.* [\[link\]](#)

18.11.2017 Decision Management Community – *Automation of Gödel's Proof of God's Existence.* [\[link\]](#)

09.11.2017 Berliner Mathematische Gesellschaft – *Berechnende Mathematik.* [\[link\]](#)

18.09.2017 Die Welt – *Wir sehen uns in der Schule* (Friederike Lübke). [\[link\]](#)

22.01.2017 CN24 – *Un calcolo matematico dimostra l'esistenza di Dio.* [\[link\]](#)

01.12.2016 Hohe Luft, Philosophie-Zeitschrift – *Christoph Benz Müller zur großen Frage.* [\[link\]](#)

08.07.2016 Bundeszentrale für politische Bildung (bpb), Berlin – *Erwachen der Roboter – Lernende Maschinen und die Intelligenz der Zukunft.* [\[link\]](#)

28.04.2016 campus.leben FU Berlin – *Forschen lernen im digitalen Zeitalter.* [\[link\]](#)

13.04.2016 Tagesspiegel.de – *FU-Lehrpreis für Christoph Benz Müller – Der Gottesbeweiser.* [\[link\]](#)

01.02.2016 Pressemitteilung FU Berlin – *Teaching Award of Freie Universität Berlin for 2015: Logic Training 3.0.*

08.08.2015 Spiegel Online – *Beweis-Software: Sind Computer die besseren Mathematiker?*

01.08.2015 Heise Online – *CADE-25: Computer-Logiker treffen sich in Berlin.*

07.10.2014 Zeit Magazin (print 18.09.2014) – *Über Gott und die Welt der Journalisten.*

22.08.2014 Zeit Online – *Math up your life!*

26.04.2014 Focus – *Warum sich Gott mathematisch nicht belegen lässt.*

31.12.2013 Telepolis – *Ranking der am meisten gelesenen und kommentierten Telepolis-Artikel 2013.*

23.12.2013 Berliner Zeitung – *Gott existiert!*

14.11.2013 United Academics, USA – *God's Existence Theorem Is Correct.*

30.10.2013 La Stampa, Italy – *Gödel e Dio: la dimostrazione adesso c'è.*

29.10.2013 HNGN, USA – *Scientists 'Prove' God's Existence with a MacBook.*

28.10.2013 La Vanguardia, Spain – *Dos científicos demuestran informáticamente la existencia de un ser superior.*

27.10.2013 abc News, USA – *Computer Scientists 'Prove' God Exists.*

27.10.2013 Delhi Daily News, India – *Researchers claim to have proven the existence of God.*

26.10.2013 Il Post, Italy – *Gödel, Dio e Repubblica.*

26.10.2013 zeenews, India – *Researchers use Apple MacBook to prove God exists.*

24.10.2013 La Repubblica, Italy – *'Sì, il teorema di Dio è corretto': due matematici fanno rivivere l'opera di Gödel.*

23.10.2013 Spiegel International – *Holy Logic: Computer Scientists 'Prove' God Exists.*

20.10.2013 myScience, Austria – *Deus Ex Machina?*

18.10.2013 Die Welt, Germany (print, front page) – *Göttliche Mathematik.*

17.10.2013 Hamburger Abendblatt – *Forscher beweisen Existenz Gottes am Computer.*

09.09.2013 Spiegel Online – *Formel von Kurt Gödel: Mathematiker bestätigen Gottesbeweis.*

02.09.2013 Frankfurter Allgemeine Zeitung, Feuilleton – *Gott lebt.*

26.08.2013 Heise Online – *Computerprogramm bestätigt Gödels Gottesbeweis.*

A more comprehensive collection of news links is available at [the GOEDELGOD press archive](#).

7 List of Publications

Citations (Google Scholar, Jun 2026): 5 948

***h*-index (Google Scholar, Jun 2026):** 39

i10-index (Google Scholar, Jun 2026): 150

Erdős number: 3 (D.M. Gabbay / S. Shelah / P. Erdős; also D.S. Scott / A. Tarski / P. Erdős)

Publications are listed below in reverse chronological order; links to preprints are provided where available.

7.1 Articles in Journals (incl. data-journal publications)

- (art1) C. Mühlenbeck and C. Benz Müller. “On the maximality of positive properties and modal collapse in variants of Gödel’s ontological proof of God”. In: *Logic and Logical Philosophy* (2026), pp. 1–21. doi: [10.12775/LLP.2026.007](https://doi.org/10.12775/LLP.2026.007). URL: <https://apcz.umk.pl/LLP/article/view/55901>.
- (art2) C. Benz Müller. “Faithful Logic Embeddings in HOL – Deep and Shallow (Isabelle/HOL dataset)”. In: *Archive of Formal Proofs (data publication)* (2025). URL: <https://www.isa-afp.org/entries/FaithfulPMLinHOL.html>.
- (art3) C. Benz Müller and D. S. Scott. “Notes on Gödel’s and Scott’s Variants of the Ontological Argument”. In: *Monatshefte für Mathematik* 208 (2025), pp. 569–611. doi: [10.1007/s00605-025-02078-x](https://doi.org/10.1007/s00605-025-02078-x).
- (art4) C. Benz Müller and D. S. Scott. “Notes on Gödel’s and Scott’s Variants of the Ontological Argument (Isabelle/HOL dataset)”. In: *Archive of Formal Proofs (data publication)* (2025). URL: https://www.isa-afp.org/entries/Notes_On_Goedels_Ontological_Argument.html.
- (art5) C. Rothgang, F. Rabe, and C. Benz Müller. “Dependently-Typed Higher-Order Logic”. In: *ACM Trans. Comput. Logic* 27.1 (2025), pp. 1–54. doi: [10.1145/3771725](https://doi.org/10.1145/3771725).
- (art6) A. Steen and C. Benz Müller. “On Reductions of Hintikka Sets for Higher-Order Logic”. In: *Journal of Applied Logics – IfCoLoG Journal of Logics and their Applications* 12.7 (2025), pp. 1813–1834. URL: <https://www.collegepublications.co.uk/downloads/ifcolog00075.pdf>.
- (art7) A. Steen, G. Sutcliffe, and C. Benz Müller. “Solving Quantified Modal Logic Problems by Translation to Classical Logics”. In: *Journal of Logic and Computation* 35.4 (2025), pp. 1–23. doi: [10.1093/logcom/exaf006](https://doi.org/10.1093/logcom/exaf006).
- (art8) J. Bayer, C. Benz Müller, K. Buzzard, M. David, L. Lamport, Y. Matiyasevich, L. Paulson, D. Schleicher, B. Stock, and E. Zelmanov. “Mathematical Proof Between Generations”. In: *Notices of the AMS* 71.1 (2024), pp. 79–92. doi: [10.1090/noti2860](https://doi.org/10.1090/noti2860). URL: <https://www.ams.org/notices/202401/rnoti-p79.pdf>.
- (art9) C. Benz Müller, D. Fuenmayor, and B. Lomfeld. “Modelling Value-oriented Legal Reasoning in LogiKey”. In: *Logics* 2.1 (2024), pp. 31–78. doi: [10.3390/logics2010003](https://doi.org/10.3390/logics2010003).
- (art10) C. Benz Müller, D. Fuenmayor, A. Steen, and G. Sutcliffe. “Who Finds the Short Proof?” In: *Logic Journal of the IGPL* 32.3 (2024), pp. 442–464. doi: [10.1093/jigpal/jzac082](https://doi.org/10.1093/jigpal/jzac082).
- (art11) X. Parent and C. Benz Müller. “Conditional Normative Reasoning as a Fragment of HOL”. In: *Journal of Applied Non-Classical Logics* 34.4 (2024), pp. 561–592. doi: [10.1080/11663081.2024.2386917](https://doi.org/10.1080/11663081.2024.2386917). URL: <https://www.tandfonline.com/doi/full/10.1080/11663081.2024.2386917>.
- (art12) X. Parent and C. Benz Müller. “Conditional Normative Reasoning as a Fragment of HOL (Isabelle/HOL dataset)”. In: *Archive of Formal Proofs (data publication)* (2024). URL: <https://www.isa-afp.org/entries/CondNormReasHOL.html>.

- (art13) A. Steen and C. Benz Müller. “Challenges for Non-Classical Reasoning in Contemporary AI Applications”. In: *Künstliche Intelligenz* 38.1-2 (2024), pp. 7–16. doi: [10.1007/s13218-024-00855-8](https://doi.org/10.1007/s13218-024-00855-8). URL: <https://link.springer.com/article/10.1007/s13218-024-00855-8>.
- (art14) A. Steen and C. Benz Müller. “Non-Classical Reasoning for Contemporary AI Applications”. In: *Künstliche Intelligenz* 38.1-2 (2024), pp. 3–5. doi: [10.1007/s13218-024-00857-6](https://doi.org/10.1007/s13218-024-00857-6). URL: <https://link.springer.com/article/10.1007/s13218-024-00857-6>.
- (art15) A. Steen and C. Benz Müller. “What are Non-Classical Logics and why do we need them? An extended interview with Dov Gabbay and Leon van der Torre”. In: *Künstliche Intelligenz* 38.1-2 (2024), pp. 17–23. doi: [10.1007/s13218-023-00824-7](https://doi.org/10.1007/s13218-023-00824-7).
- (art16) P. Baroni, C. Benz Müller, and Y. N Wáng. “Preface: Special Issue on Logic and Argumentation”. In: *Journal of Logic and Computation* 33.2 (2023), pp. 189–191. doi: [10.1093/logcom/exac091](https://doi.org/10.1093/logcom/exac091). URL: <https://doi.org/10.1093/logcom/exac091>.
- (art17) C. Benz Müller and S. Reiche. “Automating Public Announcement Logic with Relativized Common Knowledge as a Fragment of HOL in LogiKEy”. In: *Journal of Logic and Computation* 33.6 (2023), pp. 1243–1269. doi: [10.1093/logcom/exac029](https://doi.org/10.1093/logcom/exac029).
- (art18) V. Solopova, O.-I. Popescu, C. Benz Müller, and T. Landgraf. “Automated multilingual detection of Pro-Kremlin propaganda in newspapers and Telegram posts”. In: *Datenbank Spektrum* 23.1 (2023), pp. 5–14. doi: [10.1007/s13222-023-00437-2](https://doi.org/10.1007/s13222-023-00437-2). URL: <https://link.springer.com/article/10.1007/s13222-023-00437-2>.
- (art19) C. Benz Müller. “Symbolic AI and Gödel’s Ontological Argument”. In: *Zygon(r)* 57.4 (2022), pp. 953–962. doi: [10.1111/zygo.12830](https://doi.org/10.1111/zygo.12830). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/zygo.12830>.
- (art20) C. Benz Müller, D. Fuenmayor, A. Steen, and G. Sutcliffe. “Automation of Boolos’ Curious Inference in Isabelle/HOL”. In: *Archive of Formal Proofs (data publication)* 2022 (2022). URL: https://www.isa-afp.org/entries/Boolos_Curious_Inference_Automated.html.
- (art21) H. Mucha et al. “Collaborative Speculations on Future Themes for Participatory Design in Germany”. In: *i-com* 21.2 (2022), pp. 283–298. doi: [10.1515/icom-2021-0030](https://doi.org/10.1515/icom-2021-0030). URL: <https://doi.org/10.1515/icom-2021-0030>.
- (art22) C. Benz Müller. “Exploring Simplified Variants of Gödel’s Ontological Argument in Isabelle/HOL”. In: *Archive of Formal Proofs (data publication)* (2021), pp. 1–15. URL: <https://www.isa-afp.org/entries/SimplifiedOntologicalArgument.html>.
- (art23) C. Benz Müller, X. Parent, and F. Ricca. “Introduction to the Special Issue on Logic Rules and Reasoning: Selected Papers from the 2nd International Joint Conference on Rules and Reasoning (RuleML+RR 2018)”. In: *Theory and Practice of Logic Programming* 21.1 (2021), pp. 1–3. doi: [10.1017/S1471068420000113](https://doi.org/10.1017/S1471068420000113). URL: <https://doi.org/10.1017/S1471068420000113>.
- (art24) C. Benz Müller and S. Reiche. “Automating Public Announcement Logic and the Wise Men Puzzle in Isabelle/HOL”. In: *Archive of Formal Proofs (data publication)* (2021), pp. 1–5. URL: <https://www.isa-afp.org/entries/PAL.html>.
- (art25) A. Steen and C. Benz Müller. “Extensional Higher-Order Paramodulation in Leo-III”. In: *Journal of Automated Reasoning* 65.6 (2021), pp. 775–807. doi: [10.1007/s10817-021-09588-x](https://doi.org/10.1007/s10817-021-09588-x).
- (art26) C. Benz Müller, A. Farjami, D. Fuenmayor, P. Meder, X. Parent, A. Steen, L. van der Torre, and V. Zahoransky. “LogiKEy Workbench: Deontic Logics, Logic Combinations and Expressive Ethical and Legal Reasoning (Isabelle/HOL Dataset)”. In: *Data in Brief (data publication)* 33.106409 (2020), pp. 1–10. doi: [10.1016/j.dib.2020.106409](https://doi.org/10.1016/j.dib.2020.106409). URL: <https://doi.org/10.1016/j.dib.2020.106409>.

- (art27) C. Benzmüller and D. Fuenmayor. “Computer-supported Analysis of Positive Properties, Ultrafilters and Modal Collapse in Variants of Gödel’s Ontological Argument”. In: *Bulletin of the Section of Logic* 49.2 (2020), pp. 127–148. doi: [10.18778/0138-0680.2020.08](https://doi.org/10.18778/0138-0680.2020.08).
- (art28) C. Benzmüller and B. Lomfeld. “Träumen vernünftige Maschinen von Gründen? Eine reale Utopie”. In: *VerantwortungKI – Künstliche Intelligenz und gesellschaftliche Folgen (Berlin-Brandenburgische Akademie der Wissenschaften)* 3 (2020). URL: https://www.bbaw.de/files-bbaw/user_upload/publikationen/BBAW_Verantwortung-KI-3-2020_PDF-A-1b.pdf.
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7.3 Chapters in Books, Handbooks and Encyclopedias

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- (chap7) L. Höper, C. Schulte, and C. Benz Müller. “Verantwortung”. In: *Künstliche Intelligenz für Lehrkräfte – Eine fachliche Einführung mit didaktischen Hinweisen*. Ed. by U. Furbach, E. Kitzelmann, T. Michaeli, and U. Schmid. Wiesbaden: Springer Vieweg, 2024, pp. 219–236. doi: [10.1007/978-3-658-44248-4_16](https://doi.org/10.1007/978-3-658-44248-4_16). URL: https://link.springer.com/chapter/10.1007/978-3-658-44248-4_16.
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- (chap12) D. Fuenmayor and C. Benz Müller. “A Case Study On Computational Hermeneutics: E. J. Lowe’s Modal Ontological Argument”. In: *Beyond Faith and Rationality: Essays on Logic, Religion and Philosophy*. Ed. by R. Silvestre, B. Göcke, J.-Y. Béziau, and P. Bilimoria. Vol. 34. Sophia Studies in Cross-cultural Philosophy of Traditions and Cultures. Springer Nature Switzerland AG, 2020. Chap. 12. doi: [10.1007/978-3-030-43535-6_12](https://doi.org/10.1007/978-3-030-43535-6_12). URL: <https://orbilu.uni.lu/bitstream/10993/37444/1/J38-8.pdf>.

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- (chap21) S. Autexier, C. Benz Müller, D. Dietrich, and J. Siekmann. “OMEGA: Resource-Adaptive Processes in an Automated Reasoning Systems”. In: *Resource-Adaptive Cognitive Processes*. Ed. by M. W. Crocker and J. Siekmann. Cognitive Technologies. Springer, 2010, pp. 389–423. DOI: [10.1007/978-3-540-89408-7_17](https://doi.org/10.1007/978-3-540-89408-7_17). URL: <http://christoph-benzmueller.de/papers/B10.pdf>.
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- (chap27) C. Benz Müller and C. Brown. “The curious inference of Boolos in MIZAR and OMEGA”. In: *From Insight to Proof – Festschrift in Honour of Andrzej Trybulec*. Ed. by R. Matuszewski and A. Zalewska. Vol. 10(23). Studies in Logic, Grammar, and Rhetoric. The University of Białystok, Polen, 2007, pp. 299–388. URL: <http://christoph-benzmueller.de/papers/B6.pdf>.
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- (chap30) C. Benz Müller, A. Meier, and V. Sorge. “Bridging Theorem Proving and Mathematical Knowledge Retrieval”. In: *Mechanizing Mathematical Reasoning: Essays in Honor of Jörg Siekmann on the Occasion of His 60th Birthday*. Ed. by D. Hutter and W. Stephan. Vol. 2605. LNCS. Springer, 2004, pp. 277–296. doi: [10.1007/978-3-540-32254-2_17](https://doi.org/10.1007/978-3-540-32254-2_17). URL: <http://christoph-benzmueller.de/papers/B2.pdf>.
- (chap31) J. Siekmann, C. Benz Müller, A. Fiedler, A. Meier, I. Normann, and M. Pollet. “Proof Development in OMEGA: The Irrationality of Square Root of 2”. In: *Thirty Five Years of Automating Mathematics*. Ed. by F. Kamareddine. Applied Logic series (28). Kluwer Academic Publishers, 2003, pp. 271–314. doi: [10.1007/978-94-017-0253-9_11](https://doi.org/10.1007/978-94-017-0253-9_11). URL: <http://christoph-benzmueller.de/papers/B1.pdf>.

7.4 Papers at Conferences & Workshops, and Conference Keynote Papers

Some of these conferences (e.g. IJCAI, KR, ECAI, CADE, IJCAR, LPAR) are considerably more competitive than many journals.

- (conf1) R. Palesha, S. Shetty, and C. Benz Müller. “Neuro-Symbolic Translation and Verification: Automating First-Order Logic Generation and Theorem Proving from Natural Language”. In: *Information System Design: AI Driven Decision Support Systems: Proceedings of 10th International Conference on Information System Design and Intelligent Applications (ISDIA 2026), Volume 5*. Ed. by V. Bhateja, Z. R. Khan, M. Simic, and J. Anguera. Lecture Notes in Networks and Systems. Springer Nature Switzerland AG, 2026.

- (conf2) L. Pasetto, C. Benz Müller, and R. Markovich. “Right on Thought, Left Private: Reasoning about Mental Privacy in LogiKey”. In: *Proceedings of the 10th International Joint Conference on Rules and Reasoning (RuleML+RR 2026)*. Ed. by D. Lanti and A. Paschke. LNCS. Springer, 2026.
- (conf3) L. Pasetto, C. Benz Müller, and R. Markovich. “Formalizing Mental Privacy in LogiKey”. In: *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026), May 25 – 29, 2026, Paphos, Cyprus*. Ed. by C. Amato, L. Dennis, V. Mascardi, and J. Thangarajah. International Foundation for Autonomous Agents and Multiagent Systems (www.ifaamas.org), 2026, pp. 3643–3645. doi: [10.65109/PTSF2244](https://doi.org/10.65109/PTSF2244).
- (conf4) J. Schuster, D. Fuenmayor, and C. Benz Müller. “Elixir meets TPTP: Bringing Automated Reasoning to the BEAM Ecosystem”. In: *Proceedings of the 10th Workshop on Practical Aspects of Automated Reasoning (PAAR 2026), July 25, 2026, Lisbon, Portugal*. Ed. by C. Nalon, M. Suda, and M. Hajdu. CEUR Workshop Proceedings (CEUR-WS.org, ISSN 1613-0073), 2026. forthcoming.
- (conf5) P. Sigloch and C. Benz Müller. “Neuro-Symbolic Verification of LLM Outputs for Data-Sensitive Domains”. In: *KI 2026: Advances in Artificial Intelligence – 46th German Conference on Artificial Intelligence, Bremen, Germany, August 11–14, 2026, Proceedings*. Ed. by D. Wolter and G. Schwalbe. Lecture Notes in Artificial Intelligence. Springer, Cham, 2026. forthcoming (preprint [10.48550/arXiv.2605.26942](https://arxiv.org/abs/10.48550/arXiv.2605.26942)).
- (conf6) R. Aggarwal, S. Kataria, S. Urolagin, and C. Benz Müller. “Automatic Reasoning-Code Generation Using NLP-Based Entity Relations Identification”. In: *Information System Design: Big Data Analytics and Data Science. Proceedings of Ninth International Conference on Information System Design and Intelligent Applications (ISDIA 2025), Volume 3*. Ed. by V. Bhateja, M. El Barachi, A. T. Azar, and D. K. Sharma. Vol. 1539. Lecture Notes in Networks and Systems. Singapore: Springer Nature Singapore, 2025, pp. 169–183. doi: [10.1007/978-981-96-9248-4_13](https://doi.org/10.1007/978-981-96-9248-4_13).
- (conf7) C. Benz Müller. “Faithful Logic Embeddings in HOL — Deep and Shallow”. In: *Automated Deduction – CADE-30 – 30th International Conference on Automated Deduction, Stuttgart, Germany, July 28-31, 2025, Proceedings*. Ed. by C. Barrett and U. Waldmann. Vol. 15943. Lecture Notes in Computer Science. Springer, 2025, pp. 280–302. doi: [10.1007/978-3-031-99984-0_16](https://doi.org/10.1007/978-3-031-99984-0_16). URL: <https://people.mpi-inf.mpg.de/~uwe/tmp/CADE-30-Proc.pdf>.
- (conf8) D. Fuenmayor and C. Benz Müller. “HOL as a Lingua Franca for Argumentative Reasoning Agents”. In: *Conference on Artificial Intelligence and Theorem Proving (AITP-2025)*. Ed. by M. R. Douglas, T. C. Hales, C. Kaliszyk, S. Schulz, and J. Urban. 2025. URL: https://aitp-conference.org/2025/abstract/AITP_2025_paper_22.pdf.
- (conf9) D. Fuenmayor and C. Benz Müller. “Searching for Wormholes in Proof-Space”. In: *Proceedings of the 14th and 15th International Workshops on the Implementation of Logics*. Ed. by K. Korovin, S. Schulz, and M. Rawson. Vol. 21. Kalpa Publications in Computing. EasyChair, 2025, pp. 15–17. doi: [10.29007/fc89](https://doi.org/10.29007/fc89). URL: <https://easychair.org/publications/paper/scqG/>.
- (conf10) L. Lawniczak and C. Benz Müller. “Logical Modalities within the European AI Act: An Analysis”. In: *ICAIL ’25: Proceedings of the Twentieth International Conference on Artificial Intelligence and Law*. Ed. by J. Maranhão. ACM, New York, NY, USA, 2025, pp. 348–353. doi: [10.1145/3769126.3769202](https://doi.org/10.1145/3769126.3769202).
- (conf11) L. Lawniczak, L. Pasetto, C. Benz Müller, X. Li, and R. Markovich. “Reasoning with Epistemic Rights and Duties: Automating a Dynamic Logic of the Right to Know in LogiKey”. In: *ECAI 2025 - 28th European Conference on Artificial Intelligence*. Ed. by I. Lynce, N. Murano, M. Vallati, S. Villata, F. Chesani, M. Milano, A. Omicini, and M. Dastani. Vol. 413. IOS Press, 2025, pp. 1623–1630. doi: [10.3233/FAIA250988](https://doi.org/10.3233/FAIA250988). URL: <https://ebooks.iospress.nl/doi/10.3233/FAIA250988>.

- (conf12) L. Pasetto and C. Benz Müller. “The Fatio Protocol for Formal Dialogue in Isabelle/HOL”. In: *Workshop on Theorem Proving and Machine Learning in the age of LLMs: SoA and Future Perspectives, Edinburgh, Scotland, UK, April 7th-8th 2025*. Ed. by E. Komendantskaya, E. Polgreen, C. Saemann, K. Stark, and M. Rawson. 2025. URL: https://europroofnet.github.io/_pages/WG5/Edinburgh25/abstracts/20.pdf.
- (conf13) L. Pasetto and C. Benz Müller. “Visualizing Kripke Models in LogiKey: the Case of SDL”. In: *Joint Proceedings of the Workshops and Doctoral Consortium of the 41st International Conference on Logic Programming, September 9–13, 2025, Rende, Italy*. Ed. by D. Azzolini et al. Vol. 4117. CEUR Workshop Proceedings (CEUR-WS.org, ISSN 1613-0073), 2025, pp. 1–9. URL: https://ceur-ws.org/Vol-4117/LPLR2025_short_1.pdf.
- (conf14) L. Pasetto, L. Yu, D. Liga, and C. Benz Müller. “Generating Mental States for the Fatio Dialogue Protocol”. In: *The Second International Conference of Bridges between Logic, Ethics and Social Sciences (BLESS), 20-22 June 2025, School of Philosophy, Shanxi University, Taiyuan, China*. Ed. by P. Kulicki, T. Jarmuzek, and F. Ju. 2025.
- (conf15) A. Tenne, A. Vestrucci, and C. Benz Müller. “XAI in Healthcare: Analysis and Evaluation of XAI Tools and Legal Liability for Neural Networks. A Case Study on Tumor Image Classification”. In: *Information System Design: Intelligent Healthcare Informatics Proceedings of Ninth International Conference on Information System Design and Intelligent Applications (ISDIA 2025), Volume 1*. Ed. by V. Bhateja, F. Oroumchia, J. Tang, and Z. Omar. Vol. 1537. Lecture Notes in Networks and Systems. Singapore: Springer Nature Singapore, 2025, pp. 423–434. doi: [10.1007/978-981-96-9242-2_31](https://doi.org/10.1007/978-981-96-9242-2_31).
- (conf16) A. Vestrucci, C. Benz Müller, and N. Evangelatos. “Symbolic Approach to Trustworthy AI: Exploration and Healthcare Case Study”. In: *Information System Design: Intelligent Healthcare Informatics Proceedings of Ninth International Conference on Information System Design and Intelligent Applications (ISDIA 2025), Volume 1*. Ed. by V. Bhateja, F. Oroumchia, J. Tang, and Z. Omar. Vol. 1537. Lecture Notes in Networks and Systems. Singapore: Springer Nature Singapore, 2025, pp. 367–379. doi: [10.1007/978-981-96-9242-2_27](https://doi.org/10.1007/978-981-96-9242-2_27).
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- (conf18) C. Mühlenbeck and C. Benz Müller. “On the maximum of positive properties and modal collapse in Gödel’s ontological argument compared to its variants”. In: *Formal Methods and Science in Philosophy V, International Conference, Book of Abstracts*. Inter-University Centre, Ul. don Frana Bulica 4, Dubrovnik, Croatia, May 2024, pp. 31–32. URL: <https://iuc.hr/file/1802>.
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8 List of Talks

The list below includes more than 100 invited presentations. Invited talks are marked as such.

- (talk1) *(Invited Presentation) Faithful Logic Embeddings in HOL: Deep and Shallow, Propositional and Quantified*. Seminar Series on „Formalisation of mathematics with interactive theorem provers”, Cambridge University, UK, Mar. 2026.
- (talk2) *(Invited Presentation) Faithful Logic Embeddings in HOL: Deep and Shallow, Propositional and Quantified*. University of Luxembourg, Luxembourg, Mar. 2026.
- (talk3) *(Invited Presentation) KI und Gottesbeweis*. Rotary Club Bamberg – Schloss Geyerwörth, Apr. 2026.
- (talk4) *(Invited Presentation) LogiKEy Methodology: Utilizing Higher-Order Theorem Proving Systems for Universal (Meta-)Logical Reasoning*. University of Luxembourg, Luxembourg, Mar. 2026.
- (talk5) *(Invited Presentation) Many Logics, One Methodology — On the Virtues of LogiKEy in Computational Metaphysics and Ontology*. Topos Institute, Berkeley, CA, USA, May 2026. URL: <https://www.youtube.com/watch?v=oi0dnGB8HMs>.
- (talk6) *(Invited Presentation) Symbolische Kontrolle für subsymbolische Künstliche Intelligenz?* Duale Hochschule Baden-Württemberg Stuttgart, Feb. 2026.
- (talk7) *(Invited Presentation) Artificial Intelligence – Opportunities and Challenges*. BITS Pilani Dubai, Dubai, UAE, Apr. 2025.
- (talk8) *(Invited Presentation) B.Sc. Künstliche Intelligenz Data Science – Neuer Studiengang an der Otto-Friedrich-Universität Bamberg*. Agentur für Arbeit Bamberg, Mar. 2025.
- (talk9) *(Invited Presentation) Das Zentrum für Innovative Anwendungen der Informatik (ZIAI) — Innovationskreisläufe zwischen den Disziplinen*. Willkommensempfang des Graduierten-zentrums TRAc und des Welcome Centers der Universität Bamberg, July 2025.
- (talk10) *(Invited Presentation) Faithful Logic Embeddings in HOL — Deep and Shallow*. DELIGHT Workshop, University of Luxembourg, May, 2025, May 2025.
- (talk11) *(Invited Presentation) How to automate higher-order logic with Michael (fast)*. Festkolloquium MiKo 60, FAU Erlangen, 2025, May 2025.
- (talk12) *(Invited Presentation) Künstliche Intelligenz—Chancen und Herausforderungen*. German International School in Dubai (DISD), Dubai, UAE, Mar. 2025.
- (talk13) *(Invited Presentation) LogiKEy methodology: Utilizing higher-order theorem proving systems for universal logical reasoning*. University of Luxembourg, May, 2025, May 2025.
- (talk14) *(Invited Presentation) ZIAI — Zentrum für Innovative Anwendungen der Informatik*. July 2025.
- (talk15) *Faithful Logic Embeddings in HOL – Deep and Shallow*. Conference on Automated Deduction, CADE 2025, Main Track, Aug. 2025.
- (talk16) *Gödel's and Scott's variants of the ontological argument in Isabelle/HOL*. International Workshop on Computational Ontology and Metaphysics, University of Bamberg, Dec. 2025.
- (talk17) *Peter Andrews – A Short Memorial as part of CADE 2025*. Conference on Automated Deduction, CADE 2025, Business Meeting, Aug. 2025.
- (talk18) *XAI in Healthcare: Analysis and Evaluation of XAI Tools and Legal Liability for Neural Networks. A Case Study on Tumor Image Classification*. ISDIA 2025 Conference, University of Wollongong, Dubai, Jan. 2025.
- (talk19) *(Invited High-End Foreign Expert Presentation) Introduction to LogiKEy (for ethical and legal reasoning)*. Zhejiang University — ZLAIRE: University of Luxembourg Joint Lab on Advanced Intelligent Systems and REasoning, Jan. 2024. URL: <https://www.zlaire.net/full?news?2402092059260110425>.

- (talk20) *(Invited High-End Foreign Expert Presentation) LogiKEy focused introduction to Isabelle/HOL.* Zhejiang University — ZLAIRE: University of Luxembourg Joint Lab on Advanced Intelligent Systems and REasoning, Jan. 2024. URL: <https://www.zlaire.net/full?news?2402092059260110425>.
- (talk21) *(Invited Presentation) AI Consciousness – From the Perspective of a Logician.* Models of Consciousness 2024, Oct. 2024. URL: <https://amcs-community.org/events/moc5-2024/>.
- (talk22) *(Invited Presentation) Comments on the Formalisation and Automation of Foundational Theories from the Point of View of LogiKEy.* Hausdorff Institute for Mathematics, Workshop “Prospects of Formalized Mathematics”, June 2024. URL: <https://www.youtube.com/watch?v=1yYldjkZJZQ>.
- (talk23) *(Invited Presentation) Experiments in Computational Metaphysics using the logic-pluralistic LogiKEy methodology.* East China Normal University, Shanghai, China, Nov. 2024.
- (talk24) *(Invited Presentation) KI Insights: Experte Prof. Dr. Christoph Benz Müller über Ethik, Innovation und die globale Perspektive auf Künstliche Intelligenz.* Jung und Alt bewegt, Digitaler Workshop, Vereinigung Deutscher Wissenschaftler (VDW), Feb. 2024. URL: <https://vdw-ev.de/bericht-jung-und-alt-bewegt-ki/>.
- (talk25) *(Invited Presentation) KI kann mehr als Muster in Daten erkennen — Über erfolgreiche Anwendungen symbolischer KI-Technologien in Philosophie, Theologie und Mathematik.* Vortragsreihe „Zwischen Mensch und Maschine: Künstliche Intelligenz im Dialog mit den Geisteswissenschaften“, Universität Leipzig, January 23, 2024, Jan. 2024. URL: <https://www.theol.uni-leipzig.de/fakultaet/veranstaltungsportal/veranstaltungsdetail/termin/vom-hilfsmittel-zur-disziplin-die-ki-von-morgen>.
- (talk26) *(Invited Presentation) KI-Systeme in Gesellschaft und Forschung: Eine Einführung und Positionierung.* Fachforum Geisteswissenschaften 2024, Hans-Seidel-Stiftung, May 2024. URL: <https://banziana.de/zwischen-1-0-und-fleisch-blut><https://vdw-ev.de/bericht-jung-und-alt-bewegt-ki/>.
- (talk27) *(Invited Presentation) Utilizing proof assistant systems and the LogiKEy methodology for experiments on free logic in HOL.* ExtenDD Seminar, University of Lodz, Poland, Jan. 2024.
- (talk28) *(Invited Keynote) Reasonable, Trusted AI through Symbolic Ethico-legal Control and Reflection?* FLAIRS-36 Conference, Florida Artificial Intelligence Research Society, Sheraton Sandy Key Resort, Clearwater Beach, Florida, USA, May 14-17, May 2023. DOI: [10.13140/RG.2.2.32547.84003](https://doi.org/10.13140/RG.2.2.32547.84003).
- (talk29) *(Invited Presentation) Computational Metaphysics.* ZIAI-Tag Bamberg, July 2023.
- (talk30) *(Invited Presentation) Ethisch-rechtliche Kontrolle von KI Systemen.* FPS-Law Jahresempfang Berlin, June 2023.
- (talk31) *(Invited Presentation) HOL, Universal Meta-Logical Reasoning and LogiKEy.* School of Computing, Ulster University, UK, Mar. 2023.
- (talk32) *(Invited Presentation) Is HOL (as a meta-logic) all we need for the implementation of flexible normative reasoning?* Dagstuhl Seminar 23151 on Normative Reasoning for AI, Apr. 2023.
- (talk33) *(Invited Presentation) Reasonable, trusted AI requires hybrid techniques.* International Conference on Computational Intelligence and Network Systems (CINS 2023), October 18, 2023, Dubai, UAE, Oct. 2023.
- (talk34) *(Invited Presentation) Was kann KI (noch besser machen)?* Bamberg Startups e.V., Apr. 2023.
- (talk35) *(Invited Presentation) Was kann KI? Perspektiven aus der Forschung an der Universität Bamberg.* IT-Forum Oberfranken, Mar. 2023.
- (talk36) *(Invited Presentation) Who finds the short proof?* Deduktionstreffen, October 26, 2023, Berlin, Sept. 2023.

- (talk37) *ChatGPT, KI & Gott*. Servus KI! Podcast, Feb. 2023.
- (talk38) *Ethische KIs*. Servus KI! Podcast, Mar. 2023.
- (talk39) *KI für den Menschen: Ein Dreiklang aus Beweis, Verantwortung und Dialog*. Antrittsvorlesung an der Otto-Friedrich-Universität Bamberg (13. Juni 2023), June 2023.
- (talk40) *Know-it-all Parrots in the Offing*. Dagstuhl Seminar 23151 on Normative Reasoning for AI, Apr. 2023.
- (talk41) *(Invited Keynote) Bridging Legal Argumentation and Automated Theorem Proving*. Lawgorithm: AI, Ethics and Law Seminar, University of Sao Paulo, June, June 2022. URL: <https://youtu.be/h8hTsMMeHgU?t=3700>.
- (talk42) *(Invited Presentation, Slam) Vertrauenswürdige und Verantwortungsvolle KI für die Smart City*. Smart = gut? Ethische Aspekte der Smart City, Workshop, U Bamberg, 6./7. Mai, May 2022.
- (talk43) *(Invited Presentation) Der 'Beweis'*. Christmas Lecture at University of Bamberg, Dec. 2022.
- (talk44) *(Invited Presentation) Ethico-legal Governance of AI Systems*. AI-Meetup, Dependable Intelligent Software Lab, U Bamberg, 1. April, Apr. 2022.
- (talk45) *(Invited Presentation) HOL, Universal Meta-Logical Reasoning and LogiKEy*. AuReLeE Workshop Luxembourg (<https://aurelee.net/conclusion/index.html>), July 18, doi: 10.13140/RG.2.2.11932.16003, July 2022.
- (talk46) *(Invited Presentation) Kann man KI Systeme kontrollieren? Smart = gut? Ethische Aspekte der Smart City*, Workshop, U Bamberg, 6./7. Mai, May 2022.
- (talk47) *(Invited Presentation) KI an der Leine*. Acatech Science Slam Bamberg, Oct. 2022.
- (talk48) *(Invited Presentation) Logico-Pluralistic Exploration of Foundational Theories with Computers*. Topos Institute, Berkeley, CA, 16 June, June 2022. URL: <https://www.youtube.com/watch?v=c3qfeSmhN-8>.
- (talk49) *(Invited Presentation) Studying Ontological Arguments with Computers*. The Logic and Religion Webinar, 12 May, May 2022. URL: <https://www.youtube.com/watch?v=4zwWXhS7onw>.
- (talk50) *(Invited Presentation) The Computer as Experimental Metaphysician? The Humanities, The Rise of AI – Implications of Cultural, and Societal Engineering*, World Expo Dubai, February 7, Feb. 2022. URL: <https://youtu.be/X4GYd22rV0A>.
- (talk51) *(Invited Presentation) Trusted AI through Ethico-Legal Governors? Public Discussion: Digitization, Artificial Intelligence and Responsibility of Science*, Summer School, Vereinigung Deutscher Wissenschaftler, 30 June, June 2022.
- (talk52) *(Invited Presentation) Trusted AI through Ethico-Legal Governors? Summer School AI & Ethics*, Goethe-Institut e.V., Sept. 2022. URL: <https://youtu.be/XbHZgxzggkQ?feature=shared&t=1018>.
- (talk53) *(Invited Presentation) What is AI?* Summer School AI & Ethics, Goethe-Institut e.V., Sept. 2022.
- (talk54) *(Interview) Higher-Order Logic*. Ontology Talk by Adam Pease, July 2021.
- (talk55) *(Invited Inauguration Speech) Why we need ZLAIRE*. Inauguration Ceremony, Kick-off Meeting of the Zhejiang University–University of Luxembourg Joint Lab on Advanced Intelligent Systems, and REasoning (ZLAIRE), Nov. 2021. doi: 10.13140/RG.2.2.19484.62084.
- (talk56) *(Invited Presentation and Demo) Value-Oriented Ethico-Legal Reasoning in LogiKEy*. AAAI Spring Symposium on Implementing AI Ethics, Mar. 2021.
- (talk57) *(Invited Presentation) A Simplified Variant of Gödel's Ontological Argument*. Seminar in Logic and Philosophy of Mathematics, University of Helsinki, Nov. 2021.

- (talk58) *(Invited Presentation) Analysis and exploration of theistic arguments with theorem provers and model finders: New insights into the Ontological Argument.* European Academy of Religion (EuARe), Panel on Religion, Technology, and Artificial Intelligence, Aug. 2021.
- (talk59) *(Invited Presentation) Classical Higher-Order Logic (HOL)—Automation and Selected Application.* National Colloquium on AI, Mexico, Jan. 2021. URL: <https://www.youtube.com/watch?v=XaxCeVOMfQ4>.
- (talk60) *(Invited Presentation) Higher-order Free Logic and Free Logic in Higher-order Logic.* Workshop on Existence and Free Logic, Oct. 2021.
- (talk61) *(Invited Presentation) Latest Extensions to LogiKEy.* DELIGHT Project Kick-Off Workshop (online), University of Luxembourg, Dec. 2021.
- (talk62) *(Invited Presentation) Value-Oriented Ethico-Legal Reasoning in LogiKEy.* KI Club, FU Berlin, Apr. 2021.
- (talk63) *(Invited Tutorial, Covid-19-postponed) tba.* PhDs in Logic XII, 2021.
- (talk64) *Value-Oriented Ethico-Legal Reasoning in LogiKEy.* Joint annual meeting of the GI groups Deduction Systems and Logic in Computer Science, Mar. 2021.
- (talk65) *Vertrauenswürdige und Verantwortungsvolle KI für unsere große Stadt.* Science Slam der FU Berlin im Rahmen des berlinweiten Wissenschaftsfestivals „Wissensstadt Berlin 2021“, Aug. 2021.
- (talk66) *(Invited E-Lecture) Ethisch-rechtliche Kontrolle von KI Systemen.* Bitkom AI Research Network, bitkom.org, Feb. 2020.
- (talk67) *(Invited E-Lecture) Formales Argumentieren mit dem Computer – über Disziplinargrenzen hinaus.* Große Interdisziplinarität wagen!, Joint event of TU Berlin, FU Berlin, HU Berlin and Charité Berlin organized by Present Futures Forum Berlin, July 2020. doi: [10.13140/RG.2.2.14168.52487](https://doi.org/10.13140/RG.2.2.14168.52487).
- (talk68) *(Invited Keynote, Covid-19-postponed) Ethico-legal governance of intelligent artificial agents – Can post-hoc normative reasoning competencies prevent AI systems from going rogue?* Plenary at the International Conferences on Logic and AI, ZJU LogAI, Hangzhou, China, 2020. URL: <http://www.xixilogic.org/zjulogai/abstracts/>.
- (talk69) *(Invited Panelist) Una Europa Talk: Resistance is futile. You will be digitalized!* Falling Walls and Berlin Science Week, Nov. 2020. URL: <https://www.youtube.com/watch?v=cLrjmCb96Jc>.
- (talk70) *(Invited Pitch) Berlin Ethics Lab für verantwortliche KI und verantwortliche Mensch-Maschine-Interaktion.* Science Pitch "Exploration Projects Social Cohesion", Berlin University Alliance, Aug. 2020.
- (talk71) *(Invited Presentation and Panelist) Bitkom Forum Recht 2020, Rechtliche Aspekte von Künstlicher Intelligenz (KI).* Bundesverband Informationswirtschaft, Telekommunikation und Neue Medien e.V., Oct. 2020.
- (talk72) *(Invited Public Talk) Ontologischer Gottesbeweis am Computer.* Akademie Franz Hitze Haus, Münster, 6. Oktober, Oct. 2020.
- (talk73) *(Invited Seminar) The Hermeneutics of AI Interpretability.* University of Bern, Philosophy, Oct. 2020.
- (talk74) *(Invited Tutorial) Coding Ethics: Computational Techniques for Ethical Decisions.* EPFL-FU Digital Humanities Winter School "AI Ethics", EPFL Lausanne, 4-5 February, 2020, Feb. 2020.
- (talk75) *A (Simplified) Supreme Being Necessarily Exists, says the Computer: Computationally Explored Variants of Gödel's Ontological Argument.* 17th International Conference on Principles of Knowledge Representation and Reasoning (KR 2020), Sept. 2020. doi: [10.13140/RG.2.2.35270.29765](https://doi.org/10.13140/RG.2.2.35270.29765).

- (talk76) *Encoding Legal Balancing: Automating an Abstract Ethico-Legal Value Ontology in Preference Logic*. The 1st Workshop on Models of Legal Reasoning MLR 2020, hosted by the 17th International Conference on Principles of Knowledge Representation and Reasoning (KR 2020), Sept. 2020.
- (talk77) *Encoding Legal Balancing: Automating an Abstract Ethico-Legal Value Ontology in Preference Logic*. Workshop on Deontic Reasoning, University of Luxembourg, Oct. 2020. URL: <https://www.youtube.com/watch?v=drIQKmBFPgY&feature=youtu.be>.
- (talk78) *Reasonable Machines: A Research Manifesto*. 43rd German Conference on Artificial Intelligence, KI2020, September 21–25, 2020: Bamberg, Germany, Sept. 2020.
- (talk79) *(Interview) Artificial Intelligence Welcomes Humans*. Deutschen Wissenschafts- und Innovationshauses São Paulo (DWIH São Paulo), Brazil, July 2019.
- (talk80) *(Invited Featured Scientist) Der Beweis*. 71. Science Slam Berlin, SO36 Berlin Kreuzberg, May 2019. URL: <https://youtu.be/rXvboBiz7co>.
- (talk81) *(Invited Keynote) Computational Metaphysics: New Insights on Gödel's Ontological Argument and Modal Collapse*. Formal Methods and Science in Philosophy III, Dubrovnik, Croatia, Apr. 2019.
- (talk82) *(Invited Panelist)*. Panel Discussion on "Artificial Intelligence", by Nature Research (Springer Nature), with Dean Sanderson (Springer Nature), Oliver Brock (TU Berlin), Dagmar Monett (HWR Berlin), Raul Rojas (FU Berlin), Henry Ford Bau, FU Berlin, Mar. 2019.
- (talk83) *(Invited Panelist)*. Panel Discussion on "Laws in LAWS", scientific involvement in the effort to ban Lethal Autonomous Weapon Systems (LAWS), organised by Future of Life Institute <http://futureoflife.org>, with Ariel Conn (FLI) and Kate Folb (Hollywood), Konferenzzentrum der Heinrich-Böll-Stiftung, Berlin, Mar. 2019.
- (talk84) *(Invited Plenary Talk) Human vs. Nonhuman – The Need for Ethical Intelligent Systems*. Maria Sibylla Merian Centre Conviviality-Inequality in Latin America (Mecila) & Goethe-Institut São Paulo, Brazil, Invitation by FU Berlin Office in São Paulo, July 2019.
- (talk85) *(Invited Speaker) Ethisch-rechtliche Kontrolle autonomer Systeme – Machbar?* Artificial Intelligence in Automotive, Holiday Inn München, Dec. 2019.
- (talk86) *(Invited Talk) Ethisch-rechtliche Kontrolle von KI Systemen*. AI Camp Wolfsburg, organisiert von Weissenberg Group und IHK Lüneburg-Wolfsburg, Oct. 2019.
- (talk87) *(Invited) Consistent Rational Argumentation in Politics (CRAP)*. Forum Experiment!, VolkswagenStiftung, Herrenhausen Palace, Hannover, Nov. 2019.
- (talk88) *(Invited) Deontic Logic and Artificial Intelligence*. Faculty of Law, University of São Paulo, Brazil, Aug. 2019.
- (talk89) *(Invited) Ethisch Intelligente Systeme*. Symposium "Künstliche Intelligenz: Made in Germany" des Deutschen Zentrums für Luft- und Raumfahrt (DLR), Berlin, June 2019.
- (talk90) *(Invited) Higher-order automated theorem proving — Why has there been such a resistance?* ANDREI-60: Automating New-era Deductive Reasoning Event In Iberia, Tbilisi, Georgia, May 2019.
- (talk91) *(Invited) On Universal Logical Reasoning and Gödel's Ontological Argument*. Center for Intelligent Systems and their Applications (CISA), Edinburgh University, School of Informatics, Scotland, June 2019.
- (talk92) *(Invited) Reasoning about Programs, Agents and Machine Ethics: A Meta-Logical Approach*. TU Vienna, Faculty of Informatics, Oct. 2019.
- (talk93) *(Invited) Studies in Computational Metaphysics & Computational (Pseudo-)Ethics*. University of Campinas (UNICAMP), Centre for Logic, Epistemology and the History of Science, Campinas, Brazil, July 2019.

- (talk94) *Ethisch Intelligente Systeme*. Tag der offenen Tür, Dahlem Center for Machine Learning and Robotics, FU Berlin, June 2019.
- (talk95) *Explicit Normative Reasoning and Machine Ethics*. Automated Reasoning: Challenges, Applications, Directions, Exemplary Achievements (ARCADE), Workshop at CADE-27, Natal, Brazil, Aug. 2019.
- (talk96) *KI und Robotik an der FU Berlin*. Kids Presentation for Kronach-Grundschule, FU Berlin, June 2019.
- (talk97) *What Kind of Ultrafilter is Gödel's God*. Kurt Gödel: Philosophical Views, Workshop at FU Berlin <http://christoph-benzmueller.de/2019-Goedel/>, Feb. 2019.
- (talk98) *(Invited Keynote) A Flexible Infrastructure for Normative Reasoning*. 14th International Conference, DEON 2018, Utrecht, The Netherlands, 3-6 July, 2018, July 2018. doi: [10.13140/RG.2.2.27972.48007](https://doi.org/10.13140/RG.2.2.27972.48007). URL: <https://lecturenet.uu.nl/Site1/Play/6e679ba9960a48c28b9e86688b57351f1d?catalog=5f595ae1-b213-456b-af3d-5248da311475>.
- (talk99) *(Invited Keynote) Can computers help to sharpen our understanding of ontological arguments?* 12th All India Students' Conference on Science and Spiritual Quest (AISSQ), IIT Bhubaneswar, India, Oct. 2018. URL: https://youtu.be/_ceTZ2jehT4.
- (talk100) *(Invited Keynote) Experiments in Universal Logical Reasoning — How to utilise ATPs and SMT solvers for the exploration of axiom systems for category theory in free logic?* 16th International Workshop on Satisfiability Modulo Theories (SMT 2018), Oxford, UK, July 2018.
- (talk101) *(Invited Lecture) From Computational Metaphysics towards Computational (Pseudo-)Ethics*. Bath IMI Thematic Semester: Algorithms, Accountability and Ethics in Machine Learning, U Bath, UK, 2018.
- (talk102) *(Invited; celebration of Martin Davis' 90th Birthday) A Deontic Logic Reasoning Infrastructure*. Special Session on the History and Philosophy of Computing (HaPoC) at the Computability in Europe (CiE) conference, Aug. 2018.
- (talk103) *(Invited) Experiments in Universal Logical Reasoning — How to utilise ATPs and SMT solvers for the exploration of axiom systems for category theory in free logic?* Institut für Theoretische Informatik, LMU München, June 2018.
- (talk104) *(Invited) Studies in Computational Metaphysics and Computational (Pseudo-)Ethics*. Colloquium Cognitive Systems University of Ulm, Nov. 2018.
- (talk105) *Computational Hermeneutics: Using Computers to Interpret Arguments*. Second Chinese Conference on Logic and Argumentation (CLAR 2018), Hangzhou, China, June 2018.
- (talk106) *Some Reflections on a Computer-aided Theory Exploration Study in Category Theory*. 3rd Conference on Artificial Intelligence and Theorem Proving (AITP 2018), Aussois, France, 2018. URL: <https://www.youtube.com/watch?v=Y6-iVbD7Jq4>.
- (talk107) *(Invited Keynote) Computational Metaphysics: The Virtues of Formal Proofs Beyond Math*. 2nd CLE Colloquium for Philosophy and History of Formal Sciences (CLE4Science), University of Campinas, Brazil, 2017.
- (talk108) *(Invited Keynote) Recent Successes with a Meta-Logical Approach to Universal Logical Reasoning*. XX Brazilian Symposium on Formal Methods (SMBF), Recife, Brazil, 2017.
- (talk109) *(Invited Keynote) What has the Mechanisation of Category Theory in Common with Proving God's Existence?* 3. BMG Tag, Berliner Mathematische Gesellschaft e.V., Berlin, 2017.
- (talk110) *(Invited Public Presentation) Calculamus!: Analyse von Kurt Gödel's Gottesbeweis mit dem Computer*. URANIA, Berlin, 2017.
- (talk111) *(Invited Tutorial) Automated Reasoning in Higher-order and Non-classical Logics*. Pontifícia Universidade Católica do Rio Grande do Sul (PUCRS), Porto Alegre, Brazil, 2017.

- (talk112) *(Invited Tutorial) Tutorial on Universal Logic Theorem Proving in HOL*. ILIAS group, University of Luxembourg, 2017.
- (talk113) *(Invited Tutorial) Universal Logical Reasoning via Semantical Embeddings in HOL*. 2nd CLE Colloquium for Philosophy and History of Formal Sciences (CLE4Science), University of Campinas, Brazil, 2017.
- (talk114) *(Invited) Automating Free Logic in HOL, with an Experimental Application in Category Theory*. Dagstuhl Seminar 17371: Deduction Beyond First-Order Logic, Dagstuhl, 2017.
- (talk115) *(Invited) Calcuemus! — Progress in Universal Logic Reasoning and Computational Metaphysics*. Alpen-Adria Universität Klagenfurt, 2017.
- (talk116) *(Invited) Computational Metaphysics: The Virtues of Formal Computer Proofs Beyond Maths and Computer Science*. S-2: The Ubiquity of Computing: historical, philosophical issues—Commission for Computing HaPoC—25th International Congress of History of Science, and Technology, Rio de Janeiro, Brazil, 2017.
- (talk117) *(Invited) Computational Metaphysics: The Virtues of Formal Proofs Beyond Math*. ILIAS Distinguished Lectures, University of Luxembourg, 2017.
- (talk118) *(Invited) Künstliche Intelligenz—Bemerkungen zur Historie und zu aktuellen Entwicklungen*. Innovationsregion Lausitz GmbH, Cottbus, 2017.
- (talk119) *(Invited) System Demonstration: Interactive and Automated Reasoning in Isabelle/HOL*. Workshop: Bettermarks und Mathematische Beweise, bettermarks GmbH, Berlin, 2017.
- (talk120) *(Invited) Universal Logic Reasoning via Shallow Semantical Embeddings*. Workshop on Rational Enterprise Architecture Logic and Reasoning (REAL), University of Luxembourg, 2017.
- (talk121) *(Invited) What has the Mechanisation of Category Theory in Common with Proving God's Existence?* Pontifícia Universidade Católica do Rio Grande do Sul (PUCRS), 40 Years of Computer Science, 2017.
- (talk122) *Automating Emendations of the Ontological Argument in Intensional Higher-Order Modal Logic*. KI 2017 - 40th German Conference on Artificial Intelligence, 25-29 September 2017 in Dortmund, Germany, 2017.
- (talk123) *Davis-Putnam-Logemann-Loveland Algrithmus*. Alpen-Adria Universität Klagenfurt, 2017.
- (talk124) *Innovative Teaching of Computational Metaphysics*. Session on Innovative & Effective Teaching in History of Science, Technology, 25th International Congress of History of Science, and Technology, Rio de Janeiro, Brazil, 2017.
- (talk125) *Universal Logic Theorem Proving via Semantical Embeddings in HOL*. International Workshop on Logic-Based Formalisms for Legal Reasoning (LBFLR 2017), 2017.
- (talk126) *(Invited Keynote) Künstliche Intelligenz—Wohin geht die Reise?* Shared Services and Outsourcing Woche (<http://www.sharedserviceswoche.de/mediacenter>), Berlin, 2016.
- (talk127) *(Invited Lecture Course) Computational Metaphysics*. Thematic trimester 'Current Issues in the Philosophy of Practice of Mathematics and Informatics', Centre International de Mathématiques et d'Informatique de Toulouse (CIMI), France, 2016.
- (talk128) *(Invited Panel) Erwachen der Roboter – lernende Maschinen und die Intelligenz der Zukunft*. Bundeszentrale für politische Bildung – bpb, Berlin, 2016.
- (talk129) *(Invited Tutorial) A Universal Logic Theorem Proving Approach*. Berkeley-Stanford Circle in Logic and Philosophy, San Francisco, USA, 2016.
- (talk130) *(Invited) A Success Story of Higher-Order Theorem Proving in Computational Metaphysics*. Logic Colloquium, University of California, Berkeley, USA, 2016.
- (talk131) *(Invited) Automatisierung von Gödel's Gottesbeweis im Computer*. Auticon GmbH, Berlin, 2016.

- (talk132) *(Invited) Computational Metaphysics*. Central Teaching Award Acceptance Speech at FU Berlin, 2016.
- (talk133) *(Invited) Computational Metaphysics: The Virtues of Formal Proofs Beyond Maths*. Colloquium of the Berlin Mathematical School (BMS Fridays), Berlin, 2016.
- (talk134) *(Invited) The Inconsistency in Gödel's Ontological Argument: An Application of Mathematical Proof Assistants in Metaphysics*. Mathematical Logic Seminar, Stanford University, USA, 2016.
- (talk135) *(Invited) Uniform Proofs via Shallow Semantic Embeddings?* Dagstuhl Seminar 15381 — Universality of Proofs, Dagstuhl, 2016.
- (talk136) *(Poster) The Inconsistency in Gödel's Ontological Argument: A Success Story for AI in Metaphysics*. Intl. Congress on Mathematical Software (ICMS 2016), 2016.
- (talk137) *An Object-Logic Explanation for the Inconsistency in Gödel's Ontological Theory*. KI 2016 - 39th German Conference on Artificial Intelligence, 26-30 September 2016 in Klagenfurt, Austria, 2016.
- (talk138) *An Object-Logic Explanation for the Inconsistency in Gödel's Ontological Theory*. 39th German Conference on Artificial Intelligence (KI), 2016.
- (talk139) *Automating Free Logic in Isabelle/HOL*. Intl. Congress on Mathematical Software (ICMS 2016), 2016.
- (talk140) *Axiomatizing Category Theory in Isabelle/HOL*. Deduktionstreffen, 26 September 2016 in Klagenfurt, Austria, 2016.
- (talk141) *Axiomatizing Category Theory in Free Logic*. Deduktionstreffen 2016, 2016.
- (talk142) *Einsatz von Theorembeweisern in der Lehre*. Hochschuldidaktik der Informatik: 7. Fachtagung des GI-Fachbereichs Informatik und Ausbildung/Didaktik der Informatik; 13.-14. September 2016 an der Universität Potsdam, 2016.
- (talk143) *Hands-on Higher-order Modal Logics in Isabelle/HOL*. UITP 2016, Coimbra, Portugal, 2016.
- (talk144) *The OEB Plenary Debate 2016*. The global, cross-sector conference, exhibition on technology supported learning, and training (OEB), 2016. URL: <https://www.youtube.com/watch?v=ytnD-3q8ays>.
- (talk145) *TPTP and Beyond: Representation of Quantified Non-Classical Logics*. ARQNL 2016, Coimbra, Portugal, 2016.
- (talk146) *Tutorial on Reasoning in Expressive Non-Classical Logics with Isabelle/HOL*. Second Global Conference on Artificial Intelligence (GCAI), Berlin, 2016.
- (talk147) *(Invited Keynote) Experiments in Computational Metaphysics*. 9th All India Students' Conference on Science and Spiritual Quest (AISSQ), IIT Kharagpur, India, 2015.
- (talk148) *(Invited Keynote) Gödel's Ontological Argument Revisited – Findings from a Computer-supported Analysis*. 1st World Congress on Logic and Religion, João Pessoa, Brazil, 2015.
- (talk149) *(Invited Keynote) On a (Quite) Universal Theorem Proving Approach and its Application to Metaphysics*. Tableaux 2015, Wrocław, Poland, 2015.
- (talk150) *(Invited Lecture Course) Higher-Order Modal Logics*. Logic Summer School, ANU Canberra, Australia, 2015.
- (talk151) *(Invited Lecture Course) Higher-Order Modal Logics: Automation and Applications*. The 11th Reasoning Web Summer School, 2015.
- (talk152) *(Invited) "Gottesbeweis" reloaded — Analyzing Variants of the Ontological Argument with the Computer*. Lange Nacht der Wissenschaften, Berlin, Germany, 2015.
- (talk153) *(Invited) Experiments in Computational Metaphysics*. Computational Logic Seminar (MUGS), Stanford University, USA, 2015.

- (talk154) *(Invited) Experiments in Computational Metaphysics*. SRI International, Menlo Park, USA, 2015.
- (talk155) *(Invited) Higher-Order Proofs and Models – Examples from Meta-Logical Reasoning and Metaphysics*. Dagstuhl Seminar 15381 — Deduction: Models and Proofs, 2015.
- (talk156) *(Invited) Towards Computational Metaphysics*. Universität Leipzig, 2015.
- (talk157) *(Invited) Towards Computational Metaphysics — Bridging between Mathematical Logic, Artificial Intelligence and Philosophy*. Berlin Mathematical School, lecture series on 'Bridges between mathematics and its applications', Technical University Berlin, Germany, 2015.
- (talk158) *(Invited) Towards Computational Metaphysics — Techniques and Tools for Knowledge Representation and Reasoning in Expressive Ontologies*. Friedrich-Alexander-Universität, Erlangen-Nürnberg, Germany, 2015.
- (talk159) *Der ABox-Vervollständigungsalgorithmus für ALC*. Friedrich-Alexander-Universität, Erlangen-Nürnberg, Germany, 2015.
- (talk160) *Einführung in die Komplexitätsklassen P und NP*. Universität Leipzig, 2015.
- (talk161) *(Invited Keynote) On Logic Embeddings and Gödel's God*. 22nd International Workshop on Algebraic Development Techniques, Sinaia, Romania, 2014.
- (talk162) *(Invited Keynote) On Logic Embeddings and Gödel's God*. 21. Jahrestagung der GI-Fachgruppe 'Logik in der Informatik', University of Kassel, Germany, 2014.
- (talk163) *(Invited Tutorial) Higher-Order Automated Theorem Provers*. APPA Workshop at VSL'2014, Vienna, Austria, 2014.
- (talk164) *(Invited) Gödel's God on the Computer*. Informatikkolloquium, The University of Innsbruck, Austria, 2014.
- (talk165) *(Invited) Gödel's God on the Computer*. IT University Copenhagen, Denmark, 2014.
- (talk166) *(Invited) Kurt Gödel's Gottesbeweis auf dem Computer*. Lange Nacht der Wissenschaften, Berlin, Germany, 2014.
- (talk167) *(Invited) Kurt Gödel's Gottesbeweis auf dem Computer*. Fromme Professoren Marburg, Germany, 2014.
- (talk168) *Automating Gödel's Ontological Proof of God's Existence with Higher-order Automated Theorem Provers*. 21st European Conference on Artificial Intelligence (ECAI), Prague, Czech Republic, 2014.
- (talk169) *Gödel's Proof of God's Existence*. World Congress on the Square of Opposition VI, Vatican, 2014.
- (talk170) *Gödel's Unvollständigkeitssätze*. Freie Universität Berlin, Germany, 2014.
- (talk171) *HOL based First-order Modal Logic Provers*. ARQNL Workshop at VSL'2014, Vienna, Austria, 2014.
- (talk172) *(Invited Keynote) Gödel's God on the Computer*. The 10th International Workshop on the Implementation of Logics (IWIL 2013), Stellenbosch, South Africa, 2013.
- (talk173) *A Top-down Approach to Combining Logics*. 5th International Conference on Agents and Artificial Intelligence, Barcelona, Spain, 2013.
- (talk174) *Automated Consistency Checking of Expressive Ontologies — Beware of the Wrong Interpretation of Success!* The 5th International Workshop on Acquisition, Representation and Reasoning with Contextualized Knowledge (ARCOE-LogIC 2013), Corunna, Spain, 2013.
- (talk175) *Automating Quantified Conditional Logics in HOL*. 23rd International Joint Conference on Artificial Intelligence (IJCAI 2013), Beijing, China, 2013.
- (talk176) *Automating Quantified Conditional Logics is (relatively) Easy*. Deduktionstreffen 2013, Koblenz, Germany, 2013.

- (talk177) *Cut-free calculi for challenge logics in a lazy way*. International Workshop on Algebraic Logic in Computer Science (ALCS 2013), Stellenbosch, South Africa, 2013.
- (talk178) *Formalization, Mechanization and Automation of Gödel's Proof of God's Existence*. Informatik Kolloquium, Freie Universität Berlin, Germany, 2013.
- (talk179) *HOL based First-order Modal Logic Provers*. The 19th International Conference on Logic for Programming, Artificial Intelligence and Reasoning (LPAR 2013), Stellenbosch, South Africa, 2013.
- (talk180) *HOL based Universal Reasoning*. 4th World Congress and School on Universal Logic, Rio de Janeiro, Brazil, 2013.
- (talk181) *LEO-II version 1.5*. Third International Workshop on Proof Exchange for Theorem Proving (PxTP 2013), Lake Placid, NY, USA, 2013.
- (talk182) *The Watson System – An Overview*. Freie Universität Berlin, Germany, 2013.
- (talk183) *(Invited; retirement celebration of Peter Andrews) Utilizing Higher-order Automated Theorem Provers as Universal Logic Engines*. Carnegie Mellon University, USA, 2012.
- (talk184) *(Invited) Automating Expressive Classical and Non-Classical Logics with LEO-II, Computer Science Colloquium*. Computer Science Colloquium, Freie Universität Berlin, Germany, 2012.
- (talk185) *(Invited) Konzepte und Nutzen von Lambda-Ausdrücken in Java 8 (und Scala)*. Beuth Hochschule für Technik, Berlin, Germany, 2012.
- (talk186) *(Invited) LEO-II — Eine universelle Logikmaschine*. Beuth Hochschule für Technik, Berlin, Germany, 2012.
- (talk187) *(Invited) Utilizing Church's Type Theory as a Universal Logic*. Collegium Logicum Lecture Series, Kurt Gödel Society, Vienna, Austria, 2012.
- (talk188) *Gödel's Unvollständigkeitssätze*. Freie Universität Berlin, Germany, 2012.
- (talk189) *Implementing and Evaluating Provers for First-order Modal Logics*. European Conference on Artificial Intelligence (ECAI-2012), Montpellier, France, 2012.
- (talk190) *(Invited) Automating Expressive Non-classical Logics and their Combinations in Classical Higher Order Logic*. University of Potsdam, Germany, 2011.
- (talk191) *(Invited) Der Automatische Theorembeweiser LEO-II*. BMW Group, Munich, Germany, 2011.
- (talk192) *(Invited) Intelligente Werkzeuge zur Erhebung, Bereitstellung, Analyse und Kommunikation von diversifiziertem, personalisiertem, interoperablen semantischen Wissen*. Deutsches Institut für Wirtschaftsforschung, Berlin, Germany, 2011.
- (talk193) *(Invited) Mechanisierung und Automatisierung von Kombinationen klassischer und nichtklassischer Logiken in klassischer Logik höherer Stufe*. University of Hamburg, Germany, 2011.
- (talk194) *(Invited) Mechanization and Automation of Combinations of Classical and Non-Classical Logics in Classical Higher Order Logics*. Colloquium New Trends in Computational Logic, TU Dresden, Germany, 2011.
- (talk195) *Quantified Conditional Logics are Fragments of HOL*. The International Conference on Non-classical Modal and Predicate Logics Guangzhou, China, 2011.
- (talk196) *(Invited) Adaptive Assertion-Level Proofs*. The IJCAR 2010 Workshop on Evaluation Methods for Solvers and Quality Metrics for Solutions, Edinburgh, UK, 2010.
- (talk197) *(Invited) Combining Logics in Simple Type Theory (and an Application in Ontology Reasoning)*. SRI International, Menlo Park, USA, 2010.

- (talk198) *(Invited) Three presentations at Dagstuhl Seminar 10412 QSTRLib: A Benchmark Problem Repository for Qualitative Spatial and Temporal Reasoning, Dagstuhl, Germany, October 10-13, 2010: Reasoning within and about Combinations of Logics in Simple Type Theory (talk and system demonstration), Participant Introduction, and QSTRLib Use Case: Educational Question Answering on Spatial Configurations of Countries, States, and Cities.* Dagstuhl Seminar 10412: QSTRLib, Schloss Dagstuhl, Germany, 2010.
- (talk199) *(Invited) Winner Presentation: LEO-II.* CASC-J5, Fifth International Joint Conference on Automated Reasoning (IJCAR 2010), Edinburgh, UK, 2010.
- (talk200) *Combining Logics in Simple Type Theory.* 11th International Workshop on Computational Logic in Multi-Agent Systems (CLIMA XI), Lisbon, Portugal, 2010.
- (talk201) *Ontology Archaeology: Mining a Decade of Effort on the Suggested Upper Ontology.* The ECAI-10 Workshop on Automated Reasoning about Context and Ontology Evolution (ARCOE-10), Lisbon, Portugal, 2010.
- (talk202) *Progress in Automating Higher-Order Ontology Reasoning.* Workshop on Practical Aspects of Automated Reasoning (PAAR-2010), Edinburgh, UK, 2010.
- (talk203) *Reasoning with Embedded Formulas and Modalities in SUMO.* The ECAI-10 Workshop on Automated Reasoning about Context and Ontology Evolution (ARCOE-10), Lisbon, Portugal, 2010.
- (talk204) *Sigma: An Integrated Development Environment for Logical Theory Development.* The ECAI 2010 Workshop on Intelligent Engineering Techniques for Knowledge Bases (IKBET'2010), Lisbon, Portugal, 2010.
- (talk205) *Simple Type Theory as Framework for Combining Logics.* Contest presentation at the World Congress and School on Universal Logic III (UNILOG'2010), Lisbon, Portugal, 2010.
- (talk206) *Automating Access Control Logics in Simple Type Theory with LEO-II.* 24th IFIP International Information Security Conference, Pafos, Cyprus, 2009.
- (talk207) *Automating Quantified Multimodal Logics in HOL — My very first Region Connection Calculus Prover.* AAAI 2009 Spring Symposia, Stanford, USA, 2009.
- (talk208) *Granularity-Adaptive Proof Presentation (poster).* AIED, 2009.
- (talk209) *Handover of the Festschrift to Peter B. Andrews.* CADE-22, Montreal, Canada, 2009.
- (talk210) *(Invited) Automating Access Control Logics and Multimodal Logics in the Automatic Higher-Order Theorem Prover LEO-II.* Pure and Applied Logic Seminar, Carnegie Mellon University, Pittsburgh, USA, 2008.
- (talk211) *(Invited) Automating Access Control Logics and Multimodal Logics in the Automatic Higher-Order Theorem Prover LEO-II.* Kestrel Institute, Palo Alto, USA, 2008.
- (talk212) *(Invited) Exploring Properties of Multimodal Logics with the Cooperative Automatic Higher-Order Theorem Prover LEO-II.* SRI International, Menlo Park, USA, 2008.
- (talk213) *(Invited) LEO-II — A Cooperative Automatic Theorem Prover for Classical Higher-Order Logic.* Microsoft Research, Redmond, USA, 2008.
- (talk214) *(Invited) Tool Support for Formalized Mathematics: Cooperative Higher-Order Theorem Proving with LEO-II, Tutorial Dialogues on Proofs with the DIALOG demonstrator, and the PLATO/OMEGA Proof Assistant Plug-in for TeXmacs.* Formal Mathematics Seminar, University of Bonn, Germany, 2008.
- (talk215) *Exploring Properties of Normal Multimodal Logics in Simple Type Theory with LEO-II.* Deduktionstreffen, Saarbrücken, Germany, 2008.
- (talk216) *Exploring Properties of Normal Multimodal Logics in Simple Type Theory with LEO-II.* Automated Reasoning Group Lunch Talk, The University of Cambridge, UK, 2008.

- (talk217) *LEO-II - A Cooperative Automatic Theorem Prover for Classical Higher-Order Logic (System Description)*. The 4th International Joint Conference on Automated Reasoning (IJCAR 2008), Sydney, Australia, 2008.
- (talk218) *LEO-II Demo*. ESHOL Workshop at the 4th International Joint Conference on Automated Reasoning (IJCAR 2008), Sydney, Australia, 2008.
- (talk219) *LEO-II, A Higher-Order Theorem Prover (poster)*. VSI, 2008.
- (talk220) *Some Results of the LEO-II Project*. CIAO Workshop, TU Darmstadt, Germany, 2008.
- (talk221) *THF0 — The Core TPTP Language for Classical Higher-Order Logic*. The 4th International Joint Conference on Automated Reasoning (IJCAR 2008), Sydney, Australia, 2008.
- (talk222) *(Invited Lecture Course) Semantics of Higher-Order Logics*. IT University Copenhagen, Denmark, 2007.
- (talk223) *(Invited) Challenges for Automated Theorem Proving in Classical Higher Order Logics*. The University of Edinburgh, Scotland, 2007.
- (talk224) *(Invited) Challenges for Automated Theorem Proving in Classical Higher Order Logics*. University of St. Andrews, St. Andrews, Scotland, 2007.
- (talk225) *(Invited) Challenges for Automated Theorem Proving in Classical Higher Order Logics*. Heriot-Watt University, Edinburgh, Scotland, 2007.
- (talk226) *(Invited) Effiziente Automatisierung von Logik höherer Stufe—Realisierbarer Traum oder ewiger Albtraum?* Inaugural Lecture (Privatdozent) at Saarland University, Germany, 2007.
- (talk227) *Deep Inference for Automated Proof Tutoring (poster)*. KI, 2007.
- (talk228) *LEO-II, A Higher-Order Theorem Prover (poster)*. TPHOLs, 2007.
- (talk229) *Progress Report on LEO-II: An Automatic Theorem Prover for Higher-Order Logic*. TPHOLs, Kaiserslautern, Germany, 2007.
- (talk230) *Progress Report on LEO-II: An Automatic Theorem Prover for Higher-Order Logic*. Automated Reasoning Group Lunch Talk, The University of Cambridge, UK, 2007.
- (talk231) *Term Indexing for the LEO-II Prover (poster)*. Deduktionstreffen, 2007.
- (talk232) *The LEO-II Project*. Automated Reasoning Workshop, London, UK, 2007.
- (talk233) *The LEO-II Project*. Deduktionstreffen, Koblenz, Germany, 2007.
- (talk234) *The LEO-II Project*. Omega-Ultra-Textmacs Mini Workshop, Saarbrücken, Germany, 2007.
- (talk235) *The LEO-II Project (poster)*. Deduktionstreffen, 2007.
- (talk236) *The LEO-II Project (poster)*. University of Cambridge Project Poster, 2007.
- (talk237) *(Invited Lecture Course) Semantics of Higher-Order Logics*. ESSLLI 2006, Malaga, Spain, 2006.
- (talk238) *(Invited) Classical Higher-Order Logic — Semantics, Proof Theory and Automation*. University of Potsdam, Germany, 2006.
- (talk239) *(Invited) Classical Higher-Order Logic — Theory, Applications and Problems*. Saarland University (scientific habilitation talk), Germany, 2006.
- (talk240) *(Invited) Service Oriented Architectures for Mathematics Assistance Systems*. Fachhochschule Frankfurt am Main, Germany, 2006.
- (talk241) *Cut-Simulation in Impredicative Logics*. Third International Joint Conference on Automated Reasoning (IJCAR 2006), LNAI, Seattle, USA, 2006.
- (talk242) *Dialog mit einem Beweisassistenten in natürlicher Sprache*. Ringvorlesung, FR Informatik, Universität des Saarlandes, Germany, 2006.
- (talk243) *Granularity Judgments in Proof Tutoring (poster)*. KI, 2006.

- (talk244) *Judging Granularity for Automated Mathematics Teaching*. Short paper at LPAR 2006: 13th International Conference on Logic for Programming Artificial Intelligence and Reasoning, Phnom Penh, Cambodia, 2006.
- (talk245) *Term Indexing for the LEO-II Prover*. IWIL-6 workshop at LPAR 2006: The 6th International Workshop on the Implementation of Logics, Phnom Penh, Cambodia, 2006.
- (talk246) *Underspecification in Math-DIALOG*. C-Tag, SFB 378, Saarland University, Germany, 2006.
- (talk247) *(Invited) A Structured Set of Higher-Order Problems*. Dagstuhl Seminar 05431: Deduction and Applications, Schloss Dagstuhl, Germany, 2005.
- (talk248) *(Invited) Can a Higher-Order and a First-Order Theorem Prover Cooperate?* LORIA, Nancy, France, 2005.
- (talk249) *(Invited) Logik höherer Stufe: Ein geeignetes Fundament für die Mathematik und für Formale Methoden?* TU Darmstadt, Germany, 2005.
- (talk250) *(Invited) Three Approaches for Guiding the Cooperation of Mathematical Reasoning Systems: Proof Planning, Agent-based Reasoning, and Autometa Composition of Reasoning Web Service*. QSL Theme day: Integration of deductive tools. Nancy, France, 2005.
- (talk251) *Can a Higher-Order and a First-Order Theorem Prover Cooperate?* LPAR'04 Conference, Montevideo, Uruguay, 2005.
- (talk252) *Combining Proofs of Higher-Order and First-Order Automated Theorem Provers*. LPAR-05 Workshop: Empirically Successful Automated Reasoning in Higher-Order Logic (ESHOL), Montego Bay, Jamaica, 2005.
- (talk253) *Designing a Proof GUI for non-Experts: Evaluation of an Experiment*. UITP'05 Workshop (ETAPS Satellite Workshop), Edinburgh, Scotland, 2005.
- (talk254) *DIALOG (poster)*. Begehung SFB 378, 2005.
- (talk255) *Mathematical Domain Reasoning Tasks in Tutorial Natural Language Dialog on Proofs*. Theorema-Ultra-Omega-Workshop, Saarbrücken, Germany, 2005.
- (talk256) *Mathematical Domain Reasoning Tasks in Tutorial Natural Language Dialog on Proofs*. Dream Group Reunion, Edinburgh, UK, 2005.
- (talk257) *Mathematical Domain Reasoning Tasks in Tutorial Natural Language Dialog on Proofs*. Twentieth National Conference on Artificial Intelligence (AAAI-05), Pittsburgh, Pennsylvania, USA, 2005.
- (talk258) *New Directions in the OMEGA Project*. First Saarbrücken-Nancy Workshop on Higher-order Logic and Hybrid Logics, 2005.
- (talk259) *New Directions in the OMEGA Project*. CIAO'05 Workshop, Nottingham, UK, 2005.
- (talk260) *OMEGA (poster)*. Begehung SFB 378, 2005.
- (talk261) *System Description: LEO — A Resolution based Higher-Order Theorem Prover*. LPAR-05 Workshop: Empirically Successful Automated Reasoning in Higher-Order Logic (ESHOL), Montego Bay, Jamaica, 2005.
- (talk262) *(Invited) OMEGA: A Mathematical Assistance System*. Automated Reasoning Group, Cambridge University, Cambridge, UK, 2004.
- (talk263) *Agent-oriented Proof Planning*. Evaluation of the Collaborative Research Centre SFB378 Resource-adaptive Cognitive Processes, Saarbrücken, Germany, 2004.
- (talk264) *CALCULEMUS Quo Vadis?* Special Session at the IJCAR 2004 Workshop on Computer-Supported Mathematical Theory Development together with J. Siekmann, J. Calmet, and W. Windsteiger, Cork, Ireland, 2004.
- (talk265) *DIALOG: Natural Language-based Interaction with a Mathematical Assistance System*. Deduktionstreffen, Saarbrücken, Germany, 2004.

- (talk266) *DIALOG: Natural Language-based Interaction with a Mathematical Assistance System*. OMEGA talk series, Saarbrücken, Germany, 2004.
- (talk267) *OMEGA and DIALOG*. Workshop on Logic, Proofs, and Programs, Saarbrücken, Germany, 2004.
- (talk268) *Semantics and Automation of Higher-Order Logic*, Workshop on Logic, Proofs, and Programs, Nancy, France, 2004.
- (talk269) *(Invited) OMEGA — From Proof Planning towards Mathematical Knowledge Management*. MKM Symposium 2003, Edinburgh, Scotland, 2003.
- (talk270) *A New Framework for Reasoning Agents*. IJCAI-03 Workshop on Agents and Automated Reasoning, Acapulco, Mexico, 2003.
- (talk271) *Assertion Application in Theorem Proving and Proof Planning (poster)*. IJCAI-03 Poster Presentation, Acapulco, Mexico, 2003.
- (talk272) *Bemerkungen zur Semantik und Mechanisierung von Logik hoeherer Stufe*. Deduktionstrreffen, Augsburg, Germany, 2003.
- (talk273) *CALCULEMUS - Systems for Integrated Deduction and Computation*. Mathematics on the Semantic Web, Eindhoven, The Netherlands, 2003.
- (talk274) *CALCULEMUS Midterm Review Report*. Midterm Review of the European Union Research Training Network CALCULEMUS, Saarbrücken, Germany, 2003.
- (talk275) *OMEGA*. Meeting in camera of the Special Research Centre SFB 378, Dagstuhl, Germany, 2003.
- (talk276) *OMEGA - Ein Assistenzsystem für die Mathematik*. Open day, Saarland University, Saarbrücken, Germany, 2003.
- (talk277) *Proof Development with OMEGA — Square root of 2 is Irrational*. Theorema-Omega'03 Workshop, Schloss Hagenberg, Austria, 2003.
- (talk278) *Saarland University Node Report*. Midterm Review of the European Union Research Training Network CALCULEMUS, Saarbrücken, Germany, 2003.
- (talk279) *The CALCULEMUS Research Training Network — A short Overview*. CALCULEMUS Symposium 2003, Rome, Italy, 2003.
- (talk280) *The CALCULEMUS Research Training Network — A short Overview*. First QPQ Workshop on Deductive Software (QPQ'03), CADE-19, Miami, Florida, USA, 2003.
- (talk281) *Tutorial Dialogs on Mathematical Proofs*. IJCAI-03 Workshop on Knowledge Representation and Automated Reasoning for E-Learning Systems, Acapulco, Mexico, 2003.
- (talk282) *(Invited Tutorial) From Natural Deduction to Sequent Calculus and back*. CALCULEMUS Autumn School 2002, Pisa, Italy, 2002.
- (talk283) *(Invited) Tutorial Dialog with a Mathematical Assistant System*. Computer Science Department, The University of Birmingham, UK, 2002.
- (talk284) *Agent based proof search with Indexed Formulas*. CALCULEMUS 2002, Marseille, France, 2002.
- (talk285) *Agent Based Theorem Proving (poster)*. AISB, 2002.
- (talk286) *Proof Development with OMEGA*. CADE-18, Copenhagen, Denmark, 2002.
- (talk287) *Proof Development with OMEGA: Sqrt(2) is irrational*. LPAR 2002, Tbilisi, Georgia, 2002.
- (talk288) *Reasoning Services in the MathWeb-SB for Symbolic Verification of Hybrid Systems*. VERIFY'02 Workshop at FLOC 2002, Copenhagen, Denmark, 2002.
- (talk289) *Ressource-Adaptive Proof Planning with OMEGA*. Meeting in camera of the Special Research Centre SFB 378, Wallerfangen, Germany, 2002.

- (talk290) *Tutorial Dialog with a Mathematical Assistant System*. Computer Science Department, The University of Birmingham, UK, 2002.
- (talk291) *Tutorial Dialog with a Mathematical Assistant System*. Meeting in camera of the Special Research Centre SFB 378, Wallerfangen, Germany, 2002.
- (talk292) *(Invited Keynote) An Agent-based Approach to Reasoning*. AISB'01 Convention Agents, Cognition in conjunction with 8th Workshop on Automated Reasoning: Bridging the Gap between Theory, and Practice, University of York, UK, 2001.
- (talk293) *(Invited) Agent-oriented Reasoning with O-ANTS*. Pure and Applied Logic Seminar, Carnegie Mellon University, Pittsburgh, PA, USA, 2001.
- (talk294) *(Invited) Agent-oriented Reasoning with O-ANTS*. Department of Computer Science, Cornell University, Ithaca, NY, USA, 2001.
- (talk295) *(Invited) Concurrent Resource Guided Deduction*. Theoretical Computer Science Seminar, School of Computer Science, The University of Birmingham, Birmingham, UK, 2001.
- (talk296) *(Invited) Panel member of the IJCAR 2001 Workshop Future Directions in Automated Reasoning — Problems and Ideas for a New Millennium*. IJCAR 2001, Siena, Italy, 2001.
- (talk297) *A lost proof*. TPHOLS 2001, Edinburgh, Scotland, 2001.
- (talk298) *A Lost Proof (poster)*. TPHOLS, 2001.
- (talk299) *Agent-oriented theorem proving and proof planning in OMEGA*. C++ days of SFB 378 Resource adaptive cognitive processes, Mertesdorf, Germany, 2001.
- (talk300) *Agents in OMEGA*. Meeting in camera of the OMEGA group, Saarbrücken, Germany, 2001.
- (talk301) *An Agent-oriented approach to reasoning*. CALCULEMUS Workshop 2001, Siena, Italy, 2001.
- (talk302) *DIALOG (poster)*. Begehung SFB 378, 2001.
- (talk303) *Distributed Assertion Retrieval*. First International Workshop on Mathematical Knowledge Management RISC-Linz, Schloss Hagenberg, Austria, 2001.
- (talk304) *Experiments with an Agent-oriented Reasoning System*. KI 2001, Wien, Austria, 2001.
- (talk305) *OMEGA — Ressourcenadaptives Beweisplanen*. Meeting in camera of the Special Research Division SFB378, Schloss Dagstuhl, Germany, 2001.
- (talk306) *OMEGA (poster)*. Begehung SFB 378, 2001.
- (talk307) *Proof Transformation and Expansion with a Parameterizable Inference Machine (poster)*. AISB, 2001.
- (talk308) *Tutorielle Kommunikation für ein mathematisches Assistenzsystem*. Meeting in camera of the Special Research Division SFB378, Schloss Dagstuhl, Germany, 2001.
- (talk309) *(Invited) OANTS — An Open Approach at Combining Interactive and Automated Theorem Proving*. Centre for Agent Research and Development CARD, Department of Computer Science, Manchester Metropolitan University, UK, 2000.
- (talk310) *(Invited) OMEGA, MATHWEB and Friends*. Department of Artificial Intelligence, The University of Edinburgh, Edinburgh, Scotland, 2000.
- (talk311) *(Invited) Resource Guided Concurrent Deduction with O-ANTS*. Department of Artificial Intelligence, The University of Edinburgh, Edinburgh, Scotland, 2000.
- (talk312) *(Invited) Towards Agent based Theorem Proving and Proof Planning in OMEGA*. Department of Computer Science, The University of York, UK, 2000.
- (talk313) *Agent based proof planning with O-ANTS*. Systemdemonstration at the Deduktionstreffen, Saarland University, Saarbrücken, Germany, 2000.
- (talk314) *Eine übersicht zur AG Siekmann*. Deduktionstreffen, Saarland University, Saarbrücken, Germany, 2000.

- (talk315) *OANTS — An open approach at combining Interactive and Automated Theorem Proving.* CALCULEMUS Symposium 2000, St. Andrews, Scotland, 2000.
- (talk316) *Proof Planning based on a Multi Agent Architecture?">Proof Planning based on a Multi Agent Architecture?* 9th CLAM - INKA - OMRS Workshop (CIAO), Schloss Dagstuhl, Germany, 2000.
- (talk317) *Resource Guided Concurrent Deduction.* Short talk and poster presentation at Automated Reasoning Workshop 2000, King's College, London, UK, 2000.
- (talk318) *Resource Guided Concurrent Deduction (poster).* AISB, 2000.
- (talk319) *System demonstration: OMEGA, O-ANTS, and LEO.* Department of Computer Science, The University of York, York, UK, 2000.
- (talk320) *Towards agent based proof planning.* Deduktionstreffen, Saarland University, Saarbrücken, Germany, 2000.
- (talk321) *(Invited) A two layered Agent Approach for Guiding Interactive Proofs.* Theoretical Computer Science Seminar, School of Computer Science, The University of Birmingham, UK, 1999.
- (talk322) *(Invited) Extensional Higher Order Resolution, Paramodulation and RUE-Resolution.* Theoretical Computer Science Seminar, School of Computer Science, The University of Birmingham, UK, 1999.
- (talk323) *Agent Based Mathematical Reasoning.* CALCULEMUS Workshop, Trento, Italy, 1999.
- (talk324) *Agent based Proof Planning (poster).* Poster at the 6th Workshop on Automated Reasoning: Bridging the Gap between Theory and Practice; in conjunction with AISB'99 Convention, Edinburgh, Scotland, 1999.
- (talk325) *Critical Agents Supporting Interactive Theorem Proving.* 9th Portuguese Conference on Artificial Intelligence, Evora, Portugal, 1999.
- (talk326) *Extensional Higher-Order Paramodulation and RUE-Resolution.* 16th Conference on Automated Deduction, Trento, Italy, 1999.
- (talk327) *Gleichheit und Extensionalität im automatischen Beweisen in Logik höherer Stufe.* Promotionskolloquium, Saarbrücken, Germany, 1999.
- (talk328) *Ist KI eine empirische Wissenschaft?* SAG-WAS der AG Siekmann, Schloss Dagstuhl, Germany, 1999.
- (talk329) *On Automated Higher-Theorem Proving and Henkin Completeness.* Forschungskolloquium der Studienstiftung des Deutschen Volkes, Berlin, Germany, 1999.
- (talk330) *Ressourcenadaptive Vorschlagsagenten im Interaktiven Beweisen.* Kollegiatentag im Rahmen der Herbstschule Kognitionswissenschaft, Saarbrücken, Germany, 1999.
- (talk331) *Towards Fine-Grained Proof Planning with Critical Agents (poster).* AISB, 1999.
- (talk332) *Constraint Solving in Logig Programming and Automated Theorem Proving: a comparison.* 8th International Conference on Artificial Intelligence: Methodology, Systems, Applications (AIMSA'98), Sozopol, Bulgaria, 1998.
- (talk333) *Exploiting past proof experience + Experiments in the Automatic Selection of Problem-solving Strategies.* SAG-WAS'98 der AG Siekmann, Schloss Dagstuhl, Germany, 1998.
- (talk334) *Integrating TPS with OMEGA.* Workshop Inference Mechanisms in Knowledge-Based Systems: Theory and Applications, KI'98, Bremen, Germany, 1998.
- (talk335) *Integrating TPS with OMEGA.* Oberseminar AG Siekmann, Saarland University, Saarbrücken, Germany, 1998.
- (talk336) *LEO – A Higher Order Theorem Prover.* CADE, 1998.
- (talk337) *LEO — A Higher-Order Theorem Prover.* The 15th International Conference on Automated Deduction, Lindau, Germany, 1998.

- (talk338) *Model Existence for Higher-Order Logic*. Oberseminar AG Siekmann, Saarland University, Saarbrücken, Germany, 1998.
- (talk339) *System Demonstration: Integrating TPS with OMEGA*. Workshop Inference Mechanisms in Knowledge-Based Systems: Theory and Applications, KI'98, Bremen, Germany, 1998.
- (talk340) *System Demonstration: LEO — A Higher-Order Theorem Prover*. The 15th International Conference on Automated Deduction, Lindau, Germany, 1998.
- (talk341) *System Demonstration: OMEGA — A mathematical Assistant*. Evaluation of the SFB 378, Max-Planck Institute für Informatik (MPI), Saarbrücken, Germany, 1998.
- (talk342) *System Demonstration: OMEGA — a Mathematical Assistant*. Informatikforum '98 (Max-Planck-Institut für Informatik, Deutsches Forschungszentrum für Künstliche Intelligenz und Fachbereich Informatik der Uni des Saarlandes) Saarbrücken, Germany, 1998.
- (talk343) *System Demonstration: OMEGA — a Mathematical Assistant*. Deduktionstreffen, Munich, Germany, 1998.
- (talk344) *System Demonstration: OMEGA — a Mathematical Assistant*. 8th International Conference on Artificial Intelligence: Methodology, Systems, Applications (AIMSA'98), Sozopol, Bulgaria, 1998.
- (talk345) *Embedding Extensionality in Higher-Order Resolution*. Oberseminar AG Siekmann, Saarland University, Saarbrücken, Germany, 1997.
- (talk346) *Equalizing terms by Difference Reduction and Abstraction*. Automated theorem proving seminar, Department of Mathematics, Carnegie Mellon University, Pittsburgh, USA, 1997.
- (talk347) *Extensionale Resolution höherer Stufe*. Forschungskolloquium der Studienstiftung des Deutschen Volkes, Kochl am See, Germany, 1997.
- (talk348) *Extensionale Resolution höherer Stufe*. Deduktionstreffen, Schloss Dagstuhl, Germany, 1997.
- (talk349) *LEO — Towards Higher-Order Resolution*. Automated theorem proving seminar, Department of Mathematics, Carnegie Mellon University, Pittsburgh, USA, 1997.
- (talk350) *Ein kategorieller Kalkül ebener und räumlicher Netze*. Sommerakademie der Studienstiftung des Deutschen Volkes, St. Johann, Südtirol, Italy, 1996.
- (talk351) *Higher-Order Resolution and Extensionality*. Oberseminar AG Siekmann, Saarland University, Saarbrücken, Germany, 1996.
- (talk352) *HO-Differenzreduzierung*. Oberseminar AG Siekmann, Saarland University, Saarbrücken, Germany, 1995.
- (talk353) *Towards Higher-Order Theorem Proving with equality*. Deduktionstreffen, Saarland University, Saarbrücken, Germany, 1995.

9 Third-Party Funded Projects

Several grant applications not listed here are currently under review.

9.1 Competitive research grants

- Several grant proposals are currently under review for funding.
- **FWF grant LuCi – Logics for Utilitarian Conditional Oughts** (PI Dr. Xavier Parent, TU Wien; grant doi 10.55776/PAT1007025), 412 740 €; international collaborator; 2025–ongoing.
- **DFG grant – Fairness und Effizienz in Fahrzeug-Routing-Problemen** (Dr. Martin Aleksandrov), 183 100 €; host and budget responsibility in my team at FU Berlin; 2022–2024.
- **NFDI4CS – National Research Data Infrastructure for and with Computer Science**; participant; ongoing.
- **BaCaTeC / Hightech-Agenda Bayern** – visiting scholarships in artificial intelligence for Edward Zalta and Dana Scott, 2 × 3 000 €; host; 2023.

- **BMBF** – *PetraKIP: Persönliches transparentes KI-basiertes Portfolio für die Lehrerbildung – Verbundprojekt*, 785 950 €; co-PI; 2021–2024. (Own funds received: my PhD student Veronika Solopova was employed as a core person.)
- **FNR Luxembourg** – *DELIGHT: Deontic Logics for Epistemic Rights*, ~500 000 €; non-contractual partner; 2021–2024. (No funds transferable to Germany.)
- **Volkswagen Stiftung** – *CRAP: Consistent Rational Argumentation in Politics*, 120 000 €; PI; 2018–2019.
- **DFG grant LEO-III** (BE 2501/11-1) – *Automation of Higher-Order Logic*, 276 218 €; PI; 2014–2018.
- **DFG Heisenberg** (BE 2501/9-2) – *Towards Computational Metaphysics*, ~130 000 €; PI; 2015–2017.
- **DFG Heisenberg** (BE 2501/9-1) – *Automated Reasoning in Expressive Ontologies*, ~180 000 €; PI; 2012–2015.
- **FU Berlin Central Teaching Award** – lecture concept *Computational Metaphysics*, 10 000 €; PI; 2016.
- **DFG conference support** (BE 2501/12-1) – CADE-25, Berlin, 21 600 €; 2014–2015.
- **CADE-25 grants** – FU Berlin 5 000 €; AI Journal 3 000 €; Microsoft Research 2 500 €; ECCAI 1 000 €; PI; 2014–2015.
- **DFG research grant** – *Ontology Reasoning* (BE 2501/6-1, ~90 000 €; BE 2501/8-1, ~12 000 €); PI; 2009–2011.
- **EU FP7 Marie-Curie** (PIIF-GA-2008-219982) – *International Infrastructure for the Automation of Higher-Order Logic*, with G. Sutcliffe, University of Miami; ~130 000 €; PI; 2008/2009.
- **DFG travel grant** (BE 2501/4-1) – IJCAR 2008, Australia; ~2 500 €; PI; 2008.
- **DFG SFB 378** – principal co-investigator of: OMEGA (MI 04), *Agent-oriented Proof Planning*, ~900 000 €, co-PI, 2001–2008; and DIALOG (MI 03), *NL-based Interaction with a Mathematics Assistance System*, ~900 000 €, co-PI, 2001–2008.
- **GB/DAAD ARC exchange grant** – principal co-investigator, ~7 000 €; 2004.
- **EU IST** – CALCULEMUS autumn school in Pisa; ~20 000 €; PI; 2002.
- **EU Comenius** (DE-228-PA-01-2002) – CALCULEMUS autumn school in Pisa; ~30 000 €; co-PI; 2002.
- **PhD scholarship** of the Studienstiftung des Deutschen Volkes; 1996–1998.

As project coordinator or main researcher:

- **GB EPSRC grant LEO-II** (EP/D070511/1) – *Automation of Higher-Order Logic*, with L. Paulson as PI, Cambridge University, 92 512 £; 2006–2007. (I was the researcher employed on the project.)
- **EU research training network CALCULEMUS** (HPRN-CT-2000-00102) – *Integration of Symbolic Reasoning and Symbolic Computation*; ~1 500 000 €; co-coordinator; 2001–2004. (I oversaw the entire project budget.)
- **GB EPSRC grant** (GR/M99644) – *Agent-oriented Reasoning*, with M. Kerber, Birmingham; 55 944 £; postdoc; 2000. (I was the researcher employed on the project.)

9.2 Industry projects and collaborations

- Since 2024, external advisor to Aury UG, a start-up in the mental-health sector, Berlin, Germany.
- Long-standing collaboration with Dr. Daniel Kirchner (until 2025), [Head of Solidity Development](#) at the Ethereum Foundation. Daniel is a former PhD student and member of my AISE group at U Bamberg.
- 04/2021–04/2022: contracting consultant and member of the Scientific Advisory Board, IDNI AG, Liechtenstein – smart contracts, formal verification, knowledge representation and automated reasoning.
- 10/2020–04/2021: scientific consultant with the Berlin start-up Latentine GmbH – hybrid AI, machine learning, knowledge representation and automated reasoning.
- 2018: consulting collaboration with BILLON Sp. z o.o. (Warsaw and London) on smart contracts and formal verification.
- Mentor of the *EXIST-Gründerstipendium HiPOCAMPUS: Voice-based & Case-oriented Test-Preparation Platform for Medical Oral Exams* (Bundesministerium für Wirtschaft und Energie).
- Mentor of the *EXIST-Gründerstipendium 7Markets* (Förderkennzeichen 03EGSBE435, Fördersumme 117 000 €), Bundesministerium für Wirtschaft und Energie; this stipend led to the start-up Leeway

(<https://www.leeway.tech>).

- With Dr. Hans Rüdiger Lange (Innovationsregion Lausitz GmbH, Cottbus): industrial pilot study on AI for controlling, regulating and dynamically adjusting energy production networks. Beyond standard IoT 4.0 techniques, we explored agent-based modelling, machine learning, ontologies and automated reasoning.
- Initiated collaborations between KPMG Luxembourg and U Luxembourg on *ethico-legal control of intelligent systems* in finance, which led to joint projects and project proposals at U Luxembourg.
- Approximately three years of collaboration with Dr. Adam Pease (Articulate Software, Angwin, CA, USA) on expressive ontologies for industrial applications, focusing on higher-order automated reasoning.
- Conducted my diploma project at the Fraunhofer-Institut für Biomedizinische Technik, St. Ingbert (1993–1994).

10 Lecture Courses and Teaching Activities (selection)

Together with my research group, I offer lectures, seminars and projects to students of artificial intelligence, computer science, philosophy and mathematics. I have installed and currently head the BSc programme in AI & Data Science at the University of Bamberg.

Selected topics/titles covered in recent years:

- Artificial Intelligence
- AI Systems Engineering
- Automated Theorem Proving
- Automation of Logical Reasoning
- Computational Philosophy / Metaphysics
- DevOps for AI Systems
- Epistemology and Ethics of AI
- Interactive Theorem Proving
- Knowledge Representation and Reasoning
- Isabelle for the Working Logician
- Universal Reasoning (in Philosophy, Mathematics and Computer Science)
- The Mind and the Future of AI

Detailed list with further links (see also <http://christoph-benzmueller.de/teaching.html>):

- | | |
|----------|---|
| (teach1) | C. Benz Müller. <i>Freie Universität Berlin: Courses in summer term 2026 (2 entries)</i> . Freie Universität Berlin, 2026. URL: https://tinyurl.com/4ttux546 . |
| (teach2) | C. Benz Müller. <i>University of Bamberg: Courses in summer term 2026 (7 entries)</i> . University of Bamberg, 2026. URL: https://tinyurl.com/5jb8kdmx . |
| (teach3) | C. Benz Müller. <i>University of Bamberg: Courses in winter term 2025/26 (13 entries)</i> . University of Bamberg, 2026. URL: https://univis.uni-bamberg.de/form?dsc=anew/lecture_search&founds=wiai/bereic/lehrst_4/1,/3,/30,/31,/4,/57,/9,/93,/tutori_i,/tutori_0,/uetlog,/uetlog_2,/uetlog_6&nosearch=1&anonymous=1&pers=wiai/bereic/lehrst_4/benzml&ref=main&sem=2025w . |
| (teach4) | C. Benz Müller. <i>Freie Universität Berlin: Courses in summer term 2025 (2 entries)</i> . Freie Universität Berlin, 2025. URL: https://tinyurl.com/yrvdyhhf . |
| (teach5) | C. Benz Müller. <i>University of Bamberg: Courses in summer term 2025 (9 entries)</i> . University of Bamberg, 2025. URL: https://tinyurl.com/ucttshdu . |
| (teach6) | C. Benz Müller. <i>University of Bamberg: Courses in winter term 2024/25 (6 entries)</i> . University of Bamberg, 2025. URL: http://tiny.cc/m7k3001 . |

- (teach7) C. Benz Müller. *Freie Universität Berlin: Courses in summer term 2024 (2 entries)*. Freie Universität Berlin, 2024. URL: <http://tiny.cc/c7k3001>.
- (teach8) C. Benz Müller. *University of Bamberg: Courses in summer term 2024 (8 entries)*. University of Bamberg, 2024. URL: <http://tiny.cc/f7k3001>.
- (teach9) S. Ammon, C. Benz Müller, R. Moser, R. Martin, and V. Solopova. *AISE-ETH: Ethics and Epistemology of AI*. TU Berlin, FU Berlin, BUA Berlin, University of Bamberg, 2023.
- (teach10) C. Benz Müller. *AISE-ProjPrak-UR: Universal Reasoning (in Philosophy, Mathematics and Computer Science)*. University of Bamberg, 2023.
- (teach11) C. Benz Müller. *AISE-Sem-M: Masterseminar/Oberseminar zu KI-Systementwicklung*. University of Bamberg, 2023.
- (teach12) C. Benz Müller. *AISE-Sem-M: Masterseminar/Oberseminar zu KI-Systementwicklung*. University of Bamberg, 2023.
- (teach13) C. Benz Müller and D. Fuenmayor. *AISE-Auto: Automation of First- and Higher-Order Logic*. University of Bamberg, 2023.
- (teach14) C. Benz Müller and D. Fuenmayor. *AISE-UL: Universelle Logik & Universelles Schließen*. University of Bamberg, 2023.
- (teach15) C. Benz Müller and A. Vestrucci. *AISE-Proj-M: Masterprojekt KI-Systementwicklung; AI & Ethics*. University of Bamberg, 2023.
- (teach16) D. Fuenmayor and C. Benz Müller. *AISE-Proj-B: Bachelorprojekt KI-Systementwicklung*. University of Bamberg, 2023.
- (teach17) D. Fuenmayor, D. Kirchner, and C. Benz Müller. *AISE-Proj-B: Bachelorprojekt KI-Systementwicklung; Isabelle for the Working Logician*. University of Bamberg, 2023.
- (teach18) D. Fuenmayor, D. Kirchner, and C. Benz Müller. *AISE-Sem-B: Isabelle for the Working Logician*. University of Bamberg, 2023.
- (teach19) A. Vestrucci and C. Benz Müller. *AISE-Sem-B: Computational Philosophy*. University of Bamberg, 2023.
- (teach20) E. Zalta, D. Kirchner, and C. Benz Müller. *AISE-PLM: Computational Metaphysics - Mechanizing Principia Logico-Metaphysica*. University of Bamberg, 2023.
- (teach21) M. Aleksandrov and C. Benz Müller. *AISE-Proj-B: Bachelorprojekt KI-Systementwicklung*. University of Bamberg, 2022.
- (teach22) S. Ammon, C. Benz Müller, R. Moser, M. Aleksandrov, R. Martin, and V. Solopova. *Ethics and Epistemology of AI*. TU Berlin, FU Berlin, BUA Berlin, University of Bamberg, 2022.
- (teach23) C. Benz Müller. *AISE-Sem-M: Masterseminar/Oberseminar zu KI-Systementwicklung*. University of Bamberg, 2022.
- (teach24) C. Benz Müller. *Blockseminar AISE: Automated Theorem Proving and the TPTP*. University of Bamberg, University of Luxembourg, and University of Greifswald, 2022.
- (teach25) C. Benz Müller. *Oberseminar AI Systems Engineering*. University of Bamberg, 2022.
- (teach26) C. Benz Müller and D. Fuenmayor. *AISE-UL: Universelle Logik & Universelles Schließen*. University of Bamberg, 2022.
- (teach27) C. Benz Müller, D. Fuenmayor, A. Vestrucci, and R. Martin. *AISE-ProjPrak-UR: Universal Reasoning (in Philosophy, Mathematics and Computer Science)*. University of Bamberg, 2022.
- (teach28) C. Benz Müller, U. Schmid, A. Stang, and S. Kastner. *Goethe-Institut Summer School on AI & Ethics at U Bamberg*. University of Bamberg and Goethe Institut e.V. München, 2022.
- (teach29) A. Vestrucci and C. Benz Müller. *AISE-Sem-B: Computational Philosophy*. University of Bamberg, 2022.

- (teach30) C. Benz Müller. *Ethical Foundations of AI (Ethics of AI)*. TU Berlin, FU Berlin, BUA Berlin (with Sabina Ammon, Jessica Jesus de Pinho Pinhal, TU Berlin), 2021.
- (teach31) C. Benz Müller. *Universal Logical Reasoning*. Zhejiang University, Hangzhou, China, 2020.
- (teach32) C. Benz Müller and N. Alsabab. *The Mind and the Future of AI*. Freie Universität Berlin, 2020.
- (teach33) C. Benz Müller and B. Lomfeld. *Ethical and Legal Challenges in AI and Data Science*. Freie Universität Berlin, 2020.
- (teach34) C. Benz Müller, B. Lomfeld, and M. Rohmeier. *AI Ethics*. EPFL Lausanne, 2020.
- (teach35) C. Benz Müller. *Künstliche Intelligenz (autumn block course)*. Freie Universität Berlin, 2019.
- (teach36) C. Benz Müller. *Kurt Gödel: Philosophical Views*. Freie Universität Berlin, 2019.
- (teach37) C. Benz Müller. *Normative Reasoning and Machine Ethics*. Freie Universität Berlin, 2019.
- (teach38) C. Benz Müller. *Universal Logical Reasoning*. Freie Universität Berlin, 2019.
- (teach39) C. Benz Müller and C. Schommer. *Künstliche Intelligenz*. Freie Universität Berlin, 2019.
- (teach40) T. Gleissner and C. Benz Müller. *Softwareprojekt: Logik-Software*. Freie Universität Berlin, 2019.
- (teach41) C. Benz Müller. *From computational metaphysics towards computational pseudo-ethics*. Bath IMI Thematic Semester (U Bath, UK): Algorithms, Accountability and Ethics in Machine Learning, 2018.
- (teach42) C. Benz Müller and C. Schommer. *Künstliche Intelligenz*. Freie Universität Berlin, 2018.
- (teach43) C. Benz Müller. (Invited Tutorial) *Automated Reasoning in Higher-order and Non-classical Logics*. Pontifícia Universidade Católica do Rio Grande do Sul (PUCRS), Porto Alegre, Brasil, 2017.
- (teach44) C. Benz Müller. (Invited Tutorial) *Tutorial on Universal Logic Theorem Proving in HOL*. ILIAS group, University of Luxembourg, 2017.
- (teach45) C. Benz Müller. (Invited Tutorial) *Universal Logical Reasoning via Semantical Embeddings in HOL*. 2nd CLE Colloquium for Philosophy and History of Formal Sciences (CLE4Science), University of Campinas, Brasil, 2017.
- (teach46) C. Benz Müller. *KI-gestützte Pflege einer internationalen Schuldatenbank*. Freie Universität Berlin, 2017.
- (teach47) C. Benz Müller. *WYSIWYG-Editor for Higher Order Logic (Projektseminar)*. Freie Universität Berlin, 2017.
- (teach48) C. Benz Müller and R. Rojas. *Künstliche Intelligenz*. Freie Universität Berlin, 2017.
- (teach49) C. Benz Müller and C. Ullrich. *Intelligente Lehr- und Lernsysteme*. Freie Universität Berlin, 2017.
- (teach50) C. Benz Müller. (Central teaching award of FU Berlin) *Computational Metaphysics*. Freie Universität Berlin, 2016.
- (teach51) C. Benz Müller. (Invited lecture course) *Computational Metaphysics*. Thematic trimester 'Current Issues in the Philosophy of Practice of Mathematics & Informatics', Centre International de Mathématiques et d'Informatique de Toulouse (CIMI), France. June, 2016.
- (teach52) C. Benz Müller. (Invited Tutorial) *Tutorial on Reasoning in Expressive Non-Classical Logics with Isabelle/HOL*. Second Global Conference on Artificial Intelligence (GCAI), Berlin, 2016.
- (teach53) C. Benz Müller and R. Rojas. *Künstliche Intelligenz*. Freie Universität Berlin, 2016.
- (teach54) C. B. February. (Invited tutorial) *A Universal Logic Theorem Proving Approach*. Berkeley-Stanford Circle in Logic and Philosophy, San Francisco, USA, 2016.
- (teach55) C. Benz Müller. *Expressive Logiken – Theorie, Mechanisierung, Anwendungen*. Freie Universität Berlin, 2015.

- (teach56) C. Benz Müller, P. Podlech, A. Steen, and M. Wisniewski. *Entwicklung eines netzbasierten Editors zur Generierung von PDF-Dokumenten*. Freie Universität Berlin, 2015.
- (teach57) C. Benz Müller and R. Rojas. *Künstliche Intelligenz*. Freie Universität Berlin, 2015.
- (teach58) C. Benz Müller, A. Steen, and M. Wisniewski. *Logik*. Freie Universität Berlin, 2015.
- (teach59) C. Benz Müller and B. Woltzenlogel Paleo. (Invited lecture course) *Higher-Order Modal Logics: Automation and Applications*. Logic Summer School, ANU Canberra, Australia. in December, 2015.
- (teach60) C. Benz Müller and B. Woltzenlogel Paleo. (Invited tutorial) *Higher-Order Modal Logics: Automation and Applications*. Reasoning Web Summer School, 2015.
- (teach61) C. Benz Müller. *Expressive Klassische und Nichtklassische Logiken und deren Automatisierung*. Freie Universität Berlin, 2014.
- (teach62) C. Benz Müller. *Logik erster Stufe in Theorie und Praxis*. Freie Universität Berlin, 2014.
- (teach63) C. Benz Müller and R. Rojas. *Künstliche Intelligenz*. Freie Universität Berlin, 2014.
- (teach64) C. Benz Müller. *Logik erster Stufe – Theorie und Praxis*. Freie Universität Berlin, 2013.
- (teach65) R. Rojas, C. Benz Müller, and M. Block-Berlitz. *Technologien für Frage-Antwort-Systeme*. Freie Universität Berlin, 2013.
- (teach66) C. Benz Müller and G. Sutcliffe. *Working with Automated Reasoning Tools*. Saarland University, 2008.
- (teach67) S. Autexier, C. Benz Müller, and J. Siekmann. *Fortgeschrittenenpraktikum Deduktionssysteme*. Saarland University, 2007.
- (teach68) S. Autexier, C. Benz Müller, and J. Siekmann. *Oberseminar AG Siekmann*. Saarland University, 2007.
- (teach69) C. Benz Müller. (Invited lecture course) *Semantics of Higher-Order Logic*. IT University Copenhagen, 2007.
- (teach70) S. Autexier, C. Benz Müller, and C. Brown. *Mathematical Assistant Systems Shootout*. Saarland University, 2006.
- (teach71) C. Benz Müller. (Invited lecture course) *Automated Theorem Proving in Higher-Order Logics*. Technical University Darmstadt, 2006.
- (teach72) C. Benz Müller. (Invited tutorial) *Semantics of Higher-Order Logic*. ESSLLI Summer School in Logic, Language and Information, 2006.
- (teach73) C. Benz Müller and C. Brown. *Semantics and Mechanization of Classical Higher-Order Logic*. Saarland University, 2006.
- (teach74) S. Autexier, C. Benz Müller, C. Brown, A. Fiedler, and C.-P. Wirth. *Mathematical Assistant Systems Shootout*. Saarland University, 2005.
- (teach75) S. Autexier, C. Benz Müller, C. Brown, and J. Siekmann. *Artificial Intelligence*. Saarland University, 2005.
- (teach76) S. Autexier, C. Benz Müller, A. Fiedler, J. Siekmann, and C.-P. Wirth. *Mathematical Assistance Systems*. Saarland University, 2005.
- (teach77) C. Benz Müller. *Automated Theorem Proving in First-Order and Higher-Order Logic*. Saarland University, 2004.
- (teach78) C. Benz Müller, A. Fiedler, and C.-P. Wirth. *Human-Oriented Theorem Proving: Foundations and Applications*. Saarland University, 2004.
- (teach79) C. Benz Müller, A. Fiedler, and C.-P. Wirth. *Human-Oriented Theorem Proving: Foundations and Applications*. Saarland University, 2003.
- (teach80) C. Benz Müller, E. Melis, and J. Siekmann. *Artificial Intelligence*. Saarland University, 2003.

- (teach81) C. Benz Müller and J. Siekmann. *Fortgeschrittenenpraktikum Deduktionssysteme*. Saarland University, 2003.
- (teach82) C. Benz Müller and J. Siekmann. *Oberseminar AG Siekmann*. Saarland University, 2003.
- (teach83) C. Benz Müller. (Invited tutorial) *From Natural Deduction Calculus to Sequent Calculus and back*. CALCULEMUS Autumn School in Pisa, 2002.
- (teach84) C. Benz Müller. *Automated Theorem Proving in First-Order and Higher-Order Logic*. Saarland University, 2002.
- (teach85) C. Benz Müller and J. Siekmann. *Künstliche Intelligenz*. Saarland University, 2001.
- (teach86) J. Siekmann and C. Benz Müller. *Künstliche Intelligenz*. Saarland University, 1999.

11 Supervision and Mentoring (selection)

Year of completion is shown. Doctorates are highlighted in **bold**.

<i>ongoing</i>	Prof. Dr. Dr. Andrea Vestrucchi (Guest Professor in my team at U Bamberg, mentor) – U Bamberg.
<i>ongoing</i>	Dr. Tarek Besold (Habilitation , mentor / member of the <i>Fachmentorat</i>) – U Bamberg.
<i>ongoing</i>	Max Wisniewski (Doctorate , CS, Agent-based Theorem Proving, supervisor) – FU Berlin.
<i>ongoing</i>	Evgenii Shklier (Doctorate , Philosophy, co-supervisor) – U Bamberg.
<i>ongoing</i>	Anthony Peigné (Doctorate , Molecular Logics, co-supervisor) – U Rennes, France.
<i>ongoing</i>	Marc Nickert (MSc, CS, Legal Ontology & Knowledge Graphs using AI Agents, supervisor) – FU Berlin.
<i>ongoing</i>	Farzam Torbehbar (BSc, International Software Systems Science, Coalition Logics in LogiKey, supervisor) – U Bamberg.
2026	Franz Lukas Strüver (MSc, Mathematics, Computer-Assisted Verification of Kripke Models for Modal Logic, co-supervisor) – U Greifswald.
2026	Johannes Schuster (MSc, Applied Computer Science, Higher-Order Tableaux, supervisor) – U Bamberg.
2026	Mohammad Ismail Hossen (MSc, International Software Systems Science, Interactive Theorem Proving in Livebook: A Unified Interface for ATP Systems and Hybrid AI Workflows, supervisor) – U Bamberg.
2026	Afia Ibnath (MSc, CS, A Study of Real-World Structural Knowledge-Base Integration Methods in KBLAM and their Impact on LLM Response Performance, co-supervisor) – FU Berlin.
2026	Masoumeh Kordoughli Siabani (BSc, Applied Computer Science, Aufbau und Evaluation eines DevOps-Systems für das Projekt JuPL, supervisor) – U Bamberg.
2024–2025	Xiao Chi (visiting doctoral student , CS/Law, Legal Argumentation, host and co-supervisor) – Zhejiang U.
2025	Paul Sigloch (MSc, CS, Leveraging AI in Data-Sensitive Domains: A Framework for Developing Hybrid AI Systems with a Focus on Ensuring Information Integrity, supervisor) – U Bamberg.
2025	Eric Lifante (BSc, CS, Legal Reasoning in Isabelle/HOL, co-supervisor) – TU Berlin.
2025	Patrick Salvatore Stössl (MSc, Computational Humanities, Symbolic AI in Healthcare, co-supervisor) – U Bamberg.
2025	David Fuenmayor (Doctorate , CS, A Universal Mathematical Language for Argumentative Reasoning Agents, supervisor) – FU Berlin.

2025	Mefkure Hilal Yaman-Nevruz (MSc, Computational Humanities, AI Act & Norms, supervisor) – U Bamberg.
2025	Philipp Wälde (MSc, CS, Encoding of Value-Ontologies using LogiKEy, supervisor) – FU Berlin.
2024	Dr. Mennatullah Hendawi (PostDoc in my team at U Bamberg, mentor) – U Bamberg.
2024	Dr. Martin Aleksandrov (PostDoc, DFG Eigene Stelle in my team at FU Berlin, mentor) – FU Berlin.
2024	Veronika Solopova (Doctorate, summa cum laude , CS, Hybrid AI Systems in Automated Content Moderation and Analysis, supervisor) – FU Berlin.
2024	Rosae Martín Peña (Doctorate , Philosophy, Ethical Decision-Making in Trustworthy Human-AI Interaction: Linking Counterfactual Explanations and Regret, main supervisor) – U Valladolid (executed at FU Berlin & U Bamberg).
2024	Lara Lawnicak (MSc, Computational Humanities, AI & Ethics, supervisor) – U Bamberg.
2024	Alina Tenne (MSc, Computational Humanities, AI & Ethics, co-supervisor) – U Bamberg.
2024	Dr. Cordelia Mühlenbeck (BSc, Mathematics & Philosophy, Computational Philosophy, supervisor) – FU Berlin.
2024	Jule Wittkat (BSc, Mathematics, Design and Implementation of Arithmetic in the Automated Theorem Prover Leo-III, co-supervisor) – U Greifswald.
2023	Jonas Bayer (MSc, Mathematics, Syntactic Similarity in Human-Oriented ATP, supervisor) – FU Berlin.
2023	Sheri Mahender Reddy (MSc, AI, Machine Learning for Object Recognition, supervisor) – U Bamberg.
2023	Colin Rothgang (MSc, Mathematics/CS, ATP with Dependent Types, supervisor) – FU Berlin.
2022	Daniel Kirchner (Doctorate , Mathematics, Computer-Verified Foundations of Metaphysics and an Ontology of Natural Numbers in Isabelle/HOL, supervisor) – FU Berlin.
2022	Helena Lorena Winiger (MSc, Philosophy & AI, Künstliche Intelligenz, Kontext und Emotion, co-supervisor) – FU Berlin.
2022	Tomasz Tkaczyk (MSc, CS, Design and Evaluation of Embedding-Based Topic Modeling System for News Articles, co-supervisor) – FU Berlin.
2022	Nina Papenfuss (BSc, CS, Vertrauenswürdige Autonome Fahrzeuge, supervisor) – FU Berlin.
2021	Natalia Konstantinova (MSc, Mathematics, Flow Polytopes and Automated Proving, supervisor) – FU Berlin.
2021	Aleksey Gonus (MSc, Mathematics, Categorical Semantics, supervisor) – FU Berlin.
2021	Marcus Ernst (BSc, CS, Integration semantischer Sammlungsdaten, co-supervisor) – FU Berlin.
2021	Marc Nickert (BSc, CS, Test und Optimierung von LEO-III, supervisor) – FU Berlin.
2021	Jonas Bayer (BSc, Mathematics, Exploring Categories, Formally, supervisor) – FU Berlin.
2021	Dominik Blöse (BSc, CS, Legal and Ethical Control for AI Systems, supervisor) – FU Berlin.
2021	Panagiotis Markou (BSc, Mathematics, Wittgenstein’s Attack on Gödel’s Incompleteness, supervisor) – FU Berlin.
2020	Margarita Chikobava (MSc, CS, Chatbot (start-up cooperation) in Psychotherapy, supervisor) – FU Berlin.
2020	Ali Farjami (Doctorate , CS, Mechanisation of Deontic Logic, external co-supervisor) – U Luxembourg.

- 2020 Sebastian Reiche (BSc, CS, Dynamic Logic in HOL, supervisor) – FU Berlin.
- 2020 Irina Makarenko (MSc, CS, Higher-Order Free Logic, supervisor) – FU Berlin.
- 2020 David Streit (BSc, Mathematics, Mechanising Provability Logic, supervisor) – FU Berlin.
- 2020 Kim Kern (MSc, CS, Ideator Types in Electronic Brainstorming, co-supervisor) – FU Berlin.
- 2019 Tobias Gleissner (MSc, CS, Higher-Order Modal Logic, co-supervisor) – FU Berlin.
- 2019 Valeria Zahoransky (BSc, CS, Formalisation of the US Constitution, supervisor) – FU Berlin.
- 2019 Lucca Tiemens (BSc, Mathematics, Category Theory and Modeloids, supervisor) – FU Berlin.
- 2019 Markus Pohl (BSc, Politics, Artificial Intelligence in China, co-supervisor) – FU Berlin.
- 2019 Simon Robillard (**Doctorate**, CS, Automating Program Verification, committee member) – Chalmers University.
- 2019 Sebastian Böhne (**Doctorate**, CS, Theorem Proving in Education, 2nd examiner & mentor) – U Potsdam.
- 2019 Daniel Jentsch (MSc, CS, Explicit-State Model Checker, co-supervisor) – FU Berlin.
- 2019 Christian Laßan (MSc, CS, Automatisierte Verifikation von Systemarchitekturen und -spezifikationen, co-supervisor) – FU Berlin.
- 2018 Paul Meder (MSc, CS, Deontic Agency and Moral Luck, co-advisor) – U Luxembourg.
- 2018 Adrian Miloradovic (BSc, Mathematics, Compact Resolution Proofs in Leo-III, supervisor) – FU Berlin.
- 2018 Alexander Steen (**Doctorate** – with distinction, CS, Extensional Paramodulation for Higher-Order Logic and its Effective Implementation in Leo-III, supervisor) – FU Berlin. *Awarded the GI Junior Fellowship in 2018.*
- 2018 Marco Ziener (MSc, CS, Mixing Automated Theorem Proving and Machine Learning, supervisor) – FU Berlin.
- 2018 David Fuenmayor (BSc, Philosophy, Computational Hermeneutics, supervisor) – FU Berlin.
- 2018 Fabian Schütz (MSc, CS, Reconstruction of NL Arguments, supervisor) – FU Berlin.
- 2018 Claude Fabre (**Doctorate**, Philosophy, Modellismus – Ein Beitrag zur Philosophie der modernen Mathematik, technical reviewer) – FU Berlin.
- 2017 Martin Riener (**Doctorate**, CS, Applications of HO Cut-Elimination, committee member) – TU Wien.
- 2017 Hans-Jörg Schurr (MSc, CS, Preprocessing in HO Reasoning – Learning from QBF, supervisor) – TU Wien.
- 2017 Daniel Kirchner (MSc, Mathematics, Mechanisation of Principia Metaphysica, supervisor) – FU Berlin.
- 2017 Arthur Werckshagen (BSc, Mathematics, Prädikaten-Dynamische Logik in HOL) – FU Berlin.
- 2017 Hanna Lachnitt (BSc, CS, Higher-Order Intuitionistic Logic Cube, supervisor) – FU Berlin. *Received the Helmut-Veith Award at TU Wien.*
- 2017 Tobias Gleissner (BSc, CS, Automation of Higher-Order Modal Logic, supervisor) – FU Berlin.
- 2017 Lauri Loiskoski (**Doctorate**, Separators of Simple Polytopes, mentor) – Freie Universität Berlin.
- 2016 Irina Makarenko (BSc, CS, Automation of Higher-Order Free Logic, supervisor) – FU Berlin.

2016	Samuel Gfrörer (BSc, CS, Parser for the HOL ATP Leo-III, supervisor) – FU Berlin.
2016	Robert Spangenberg (Doctorate , Landmark-based Localization for Autonomous Vehicles, committee member) – FU Berlin.
2015	Maximilian Claus (MSc, CS, Software Model Checking with Higher-Order ATP, supervisor) – FU Berlin.
2015	Kim Kern (BSc, CS, Improved Computation of CNF in HOL, supervisor) – FU Berlin.
2015	Benjamin Eckstein (MSc, CS, Benutzer-adaptives zeitkritisches Empfehlungssystem, supervisor) – FU Berlin.
2015	Dennis Griebach (Doctorate , Stereo-Vision-Aided Inertial Navigation, committee member) – FU Berlin.
2015	Shuiying Wang (Doctorate , State Lattice-based Motion Planning for Autonomous On-Road Driving, committee member) – FU Berlin.
2014	Thomas Rath (Doctorate , Benchmarking Non-Classical Logics, committee member) – Universität Potsdam.
2014	Alexander Steen (MSc, CS, Efficient Data Structures for ATP in HOL, supervisor) – FU Berlin.
2014	Max Wisniewski (MSc, CS, Agent-based Blackboard Architecture for ATP, supervisor) – FU Berlin.
2014	Marco Träger (Diploma, CS, Computer-assisted Generation of Invariants, committee member) – FU Berlin.
2014	Daniel Kühlwein (Doctorate , Machine Learning for Automated Reasoning, external committee member) – Radboud University Nijmegen, NL.
2014	Sven Olufs (Doctorate , committee member) – FU Berlin.
2013	Yves Müller (MSc, CS, Subprover Parallelism in LEO-II, supervisor) – FU Berlin.
2013	Masood Ghayoomi (Doctorate , From HPSG-based Persian Treebanking to Parsing: Machine Learning for Data Annotation, committee member) – FU Berlin.
2013	Roman Gilbourd (Doctorate , committee member) – FU Berlin.
2013	Jens Otten (Doctorate , Connection Calculi, committee member) – University of Potsdam.
2013	Marco Ziener (BSc, CS, Inconsistency Detection in Expressive Ontologies, supervisor) – FU Berlin.
2013	Priya Gopalan (Doctorate , CS, Multiagent Theorem Proving, committee member) – University of Edinburgh, GB.
2012	Max Wisniewski (BSc, Expression Evaluation Using an SMT-Solver, supervisor) – FU Berlin.
2011	Dominik Dietrich (Doctorate , scientific tutor) – Saarland University, SFB 378.
2010	Marvin Schiller (Doctorate , Granularity Analysis for Tutoring Mathematical Proofs, supervisor) – Saarland University, SFB 378, Studienstiftung.
2009	Yecheng Gu (BSc, co-supervisor) – Saarland University.
2008	Jürgen Zimmer (Doctorate , MathServe: A Framework for Semantic Reasoning Services, scientific tutor and co-supervisor) – Saarland University, SFB 378.
2008	Valentin Dimitrov (Diploma, co-supervisor) – Saarland University.
2008	Jonathan Osthof (BSc, co-supervisor) – Saarland University.
2007	Stefano Zacchiroli (Doctorate , committee member) – University of Bologna, Italy.
2007	Stephanie Ehrbächer (MSc, supervisor) – Saarland University, SFB 378.
2006	Dominik Dietrich (Diploma, co-supervisor) – Saarland University, SFB 378.
2006	Marc Wagner (Diploma, co-supervisor) – Saarland University, SFB 378.
2006	Robert Vollmann (BSc, co-supervisor) – Saarland University.
2005	Axel Schairer (Doctorate , scientific tutor) – Saarland University.
2005	Tim Priesnitz (Doctorate , committee member) – Saarland University.

2005 Marc Buckley (Diploma, supervisor) – Saarland University, SFB 378.

2005 Marvin Schiller (Diploma, supervisor) – Saarland University, SFB 378.

2005 Frank Theiss (Diploma, supervisor) – Saarland University, SFB 378.

2005 Stephanie Ehrbächer (BSc, supervisor) – Saarland University, SFB 378.

2005 Andreas Franke (Diploma, co-supervisor) – Saarland University, SFB 378.

2005 Malte Hübner (Diploma, supervisor) – Saarland University, SFB 378.

2001 Volker Sorge (**Doctorate**, *QANTS – A Blackboard Architecture for the Integration of Reasoning Techniques into Proof Planning*, committee member and scientific tutor) – Saarland University.

2001 Armin Fiedler (**Doctorate**, *User-adaptive Proof Explanation*, committee member) – Saarland University, SFB 378.

2000 Karsten Konrad (**Doctorate**, *Model Generation for Natural Language Interpretation and Analysis*, co-supervisor) – Saarland University, SFB 378.

1999 Stephan Hess (Diploma, co-supervisor) – Saarland University.